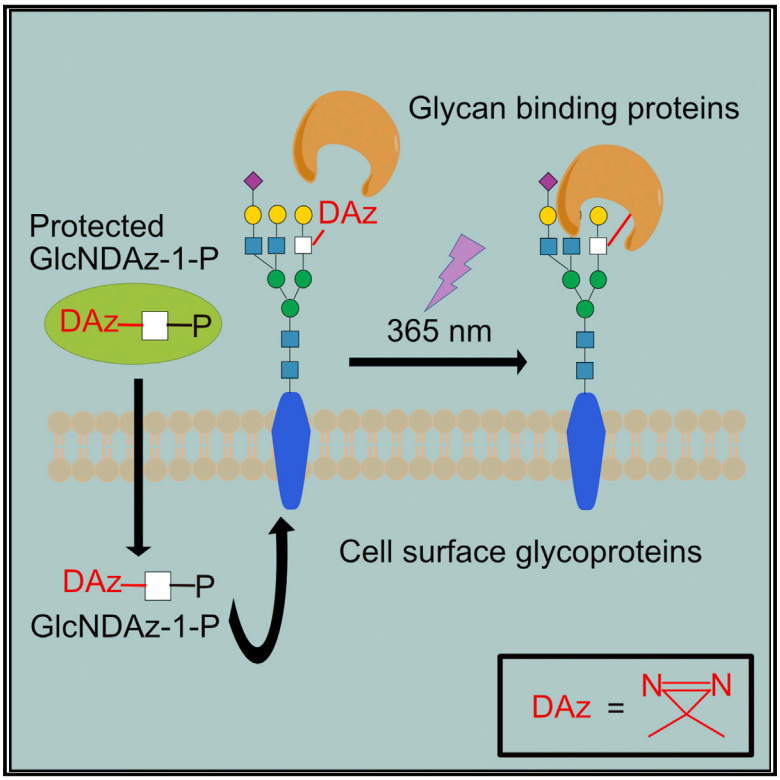
Cell Chemical Biology

A photo-cross-linking GlcNAc analog enables covalent capture of N-linked glycoprotein-binding partners on the cell surface

Graphical abstract



Highlights

- A photo-cross-linking analog of GlcNAc can be incorporated into cellular N-glycans
- Incorporation of photo-cross-linking GlcNAc modestly perturbs N-glycan structures
- Interactions between N-glycans and glycan-binding proteins are covalently captured
- In-cell cross-linking gives context-specific information about glycan binding events

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In brief

Wu et al. report that a photo-cross-linking analog of GlcNAc can be incorporated into cellular N-glycans and used to covalently cross-link glycan-protein interactions in their native cellular context. This method can be used to identify novel glycan-protein interactions and to evaluate the extent of glycan-protein binding occurring under diverse conditions.

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A photo-cross-linking GlcNAc analog enables covalent capture of N-linked glycoprotein-binding partners on the cell surface

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SUMMARY

N-glycans are displayed on cell-surface proteins and can engage in direct binding interactions with membrane-bound and secreted glycan-binding proteins (GBPs). Biochemical identification and characterization of glycan-mediated interactions is often made difficult by low binding affinities. Here we describe the metabolic introduction of a diazirine photo-cross-linker onto N-acetylglucosamine (GlcNAc) residues of N-linked glycoproteins on cell surfaces. We characterize sites at which diazirine-modified GlcNAc is incorporated, as well as modest perturbations to glycan structure. We show that diazirine-modified GlcNAc can be used to covalently cross-link two extracellular GBPs, galectin-1 and cholera toxin subunit B, to cell-surface N-linked glycoproteins. The extent of cross-linking correlates with display of the preferred glycan ligands for the GBPs. In addition, covalently cross-linked complexes could be isolated, and protein components of cross-linked N-linked glycoproteins were identified by proteomics analysis. This method may be useful in the discovery and characterization of binding interactions that depend on N-glycans.

INTRODUCTION

Cell surfaces are covered with diverse glycosylated molecules. The identity, abundance, and distribution of cell-surface glycans provide an extracellular report on cell type and physiological state. This cell-surface glycan communique is detected and interpreted by glycan-binding proteins (GBPs) that recognize specific glycan features (Varki, 2017). For example, galectins are a family of animal GBPs that share a conserved carbohydrate recognition domain and preferentially bind β-galactosides, recognizing both animal and microbial glycans (Kamili et al., 2016; Vasta, 2020). Binding of galectins to cell surfaces activates intracellular signaling pathways, but galectins are also found inside of cells, where they act directly on signaling components (Johannes et al., 2018). Acting in diverse ways, galectins play immunomodulatory roles and contribute to metastatic phenotypes (Girotti et al., 2020; Thiemann and Baum, 2016). Human cell-surface glycans are also recognized by microbial pathogens through cell-surface and secreted GBPs (Nizet et al., 2017). For instance, *Vibrio cholerae* produces a secreted toxin that causes the severe diarrhea that characterizes the disease cholera (Holmgren, 1981). In the initial step of host-cell intoxication, cholera toxin subunit B (CTB) binds glycan receptors on the surface of intestinal epithelial cells (Wernick et al., 2010). CTB binding partners include the ganglioside GM1, which CTB binds with high affinity, as well as lower-affinity fucosylated glycoconjugate ligands that are abundant on intestinal epithelial cell surfaces (Cervin et al., 2018; Holmgren et al., 1975).

Historically, affinity purification has been a powerful approach for identifying protein binding partners but, in the case of GBPs, affinity purification has limitations. First, GBPs commonly exhibit moderate to low single-site binding affinities for ligands, with micromolar and sometimes even millimolar K_d (Nizet et al., 2017). Low-affinity binding interactions dissociate rapidly and can be difficult to capture. Second, affinity purification requires lysis of cells and destruction of the cellular architecture, which can result in scrambling of binding partners. Spurious interactions may occur between molecules that would normally reside in separate compartments. Conversely, molecules that normally interact when concentrated within a specific organelle can dissociate when organellar structure is disrupted. Thus, alternate approaches have been pursued for identifying true GBP ligands.

Photoactivatable cross-linking probes (photo-cross-linkers) are tools used to capture binding interactions in normal cellular

contexts (Mishra et al., 2020; Wu and Kohler, 2019). Photo-cross-linkers can be appended to cellular biomolecules. UV irradiation converts the cross-linker to a highly reactive species that covalently reacts with nearby molecules. Cross-linked complexes can be isolated and characterized by various approaches, including immunoblotting and mass spectrometry (MS). We and others have appended photo-cross-linkers to monosaccharides and used metabolic glycan engineering (MGE) to incorporate photo-cross-linking sugars into cellular glycoconjugates (Feng et al., 2013; Han et al., 2005; Luchansky et al., 2004; Tanaka and Kohler, 2008; Yu et al., 2012). Such strategies have been used to capture and identify glycoconjugate ligands of GBPs.

We reported a method to produce a diazirine-functionalized UDP-N-acetylglucosamine (GlcNAc) donor (UDP-GlcNDAz) and showed that the nucleocytoplasmic O-GlcNAc transferase (OGT) can accept this modified donor, resulting in production of diazirine-functionalized O-GlcNAc (Yu et al., 2012). We speculated that UDP-GlcNDAz might also be a substrate for Golgi-resident glycosyltransferases, resulting in production of diazirine-modified cell-surface glycans. Evaluating whether GlcNDAz is incorporated into cell-surface glycans requires an analytic approach that provides information about sites of diazirine modification as well as overall glycan structures. First, because normal metabolic pathways interconvert sugar analogs, the possibility exists for the photo-cross-linkers to be displayed in unexpected locations. Indeed, in other MGE applications, interconversion of azide-modified monosaccharides has been observed and exploited (Boyce et al., 2011). A second consideration arises regarding sites of unnatural sugar incorporation. For example, typical N-glycans contain multiple GlcNAc residues added by different GlcNAc transferases: two GlcNAc residues in the core, GlcNAc residues initiating each mature antenna, and possibly bisecting GlcNAc and/or additional GlcNAc residues within poly-N-acetyllactosamine (polyLacNAc) chains attached to antennae. The relative ability of different glycosyltransferases to add photo-cross-linking sugars and the resulting positions at which unnatural photo-cross-linking sugars are incorporated will affect cross-linking events. Finally, incorporation of unnatural sugars has the potential to perturb glycan biosynthesis in ways that could affect glycan recognition and function. For instance, production of UDP-GlcNDAz has the potential to disrupt regulation of the hexosamine biosynthetic pathway. Also, incorporation of unnatural sugars could prevent addition of subsequent sugars, resulting in truncated glycan structures. Together, these concerns highlight the need for comprehensive analysis of unnatural sugar metabolism and incorporation for reliable use of this chemical biology tool.

Here we report that diazirine-modified GlcNAc (GlcNDAz) is incorporated into N-glycans of human cells grown in culture. The method builds on our prior demonstration that cells can be engineered to produce the diazirine-modified donor, UDP-GlcNDAz (Yu et al., 2012). We find that, like OGT, at least one Golgi-resident GlcNAc transferase, MGAT2, can use UDP-GlcNDAz as a substrate, while some Golgi-resident GlcNAc transferases do not tolerate the diazirine. Glycoproteomics analysis demonstrates production of mature complex N-glycans containing GlcNDAz and no other diazirine-modified sugars. However, lectin binding analysis does reveal modest perturbations to N-glycan structure. Using metabolically incorporated GlcNDAz, we show cross-linking of two GBPs—galectin-1 and CTB—to glycoprotein binding partners. Cross-linking intensity reflects the affinity of binding interactions. Also, galectin-1 cross-linked complexes were purified and analyzed by MS-based proteomics analysis to provide a list of candidate galectin-1 binding partners. We confirmed the cross-linking of galectin-1 to several glycoproteins, including LY75, which, to our knowledge, has not been described previously as a galectin-1 ligand. We find that the cross-linking method provides context-specific information about binding interactions: cross-linking of extracellularly added galectin-1 to LAMP1 is detected only when the cells are lysed, allowing compartmental mixing to occur. Thus, GlcNDAz cross-linking has potential application for discovery of cellular binding interactions that depend on N-linked glycosylation, and characterization of cellular conditions under which these interactions occur.

RESULTS

Strategy for metabolic production of photo-cross-linking N-glycans

Previously, we reported that HeLa (Yu et al., 2012) and K562 (Rodriguez et al., 2015) cells stably expressing the F383G mutant of UAP1 and treated with a cell-permeative, diazirine-modified analog of GlcNAc-1-phosphate (Ac_3GlcNDAz-1P(AcSATE)$_2$) produce UDP-GlcNDAz (Figure 1). We found that OGT accepts UDP-GlcNDAz in place of UDP-GlcNAc, enabling production of O-GlcNDAzylated proteins with diazirine attached to O-GlcNAc. We demonstrated the use of this metabolically incorporated cross-linker to covalently cross-link O-GlcNAcylated proteins to nearby proteins in living cells.

With a robust method for cellular production of UDP-GlcNDAz in hand, in this work we asked whether GlcNDAz was also incorporated into N-glycans. We performed experiments in T84 cells, a colorectal cancer cell line capable of producing complex-type N-glycans, including multiantennary, fucosylated, and sialylated structures (Holst et al., 2016). T84 cells stably expressing UAP1(F383G) were prepared by lentiviral infection (Rodriguez et al., 2015; Yu et al., 2012). T84 UAP1(F383G) cells cultured with Ac_3GlcNDAz-1P(AcSATE)$_2$ produced UDP-GlcNDAz efficiently, with UDP-GlcNDAz levels exceeding UDP-GlcNAc levels after 8 h (Figure S1).

GlcNDAz is incorporated into N-glycans

To incorporate GlcNDAz into N-glycans, UDP-GlcNDAz must enter the Golgi and be accepted by a Golgi-resident GlcNAc transferase (Figure 1). We assayed nine Golgi-resident GlcNAc transferases for GlcNDAz-transferase activity. Each GlcNAc transferase was incubated with an appropriate acceptor substrate (see STAR Methods) and UDP-GlcNDAz. Comparison reactions were performed with UDP-GlcNAc. Activity was quantified by measuring the release of UDP. MGAT2 showed modest activity with UDP-GlcNDAz (activity with UDP-GlcNDAz 19% of that observed with UDP-GlcNAc) and minimal activity was observed for MGAT5 (activity with UDP-GlcNDAz 4% of that observed with UDP-GlcNAc). No activity with UDP-GlcNDAz was detected for other GlcNAc transferases examined (Figures 2A and S2). MGAT2 and MGAT5 are involved in the

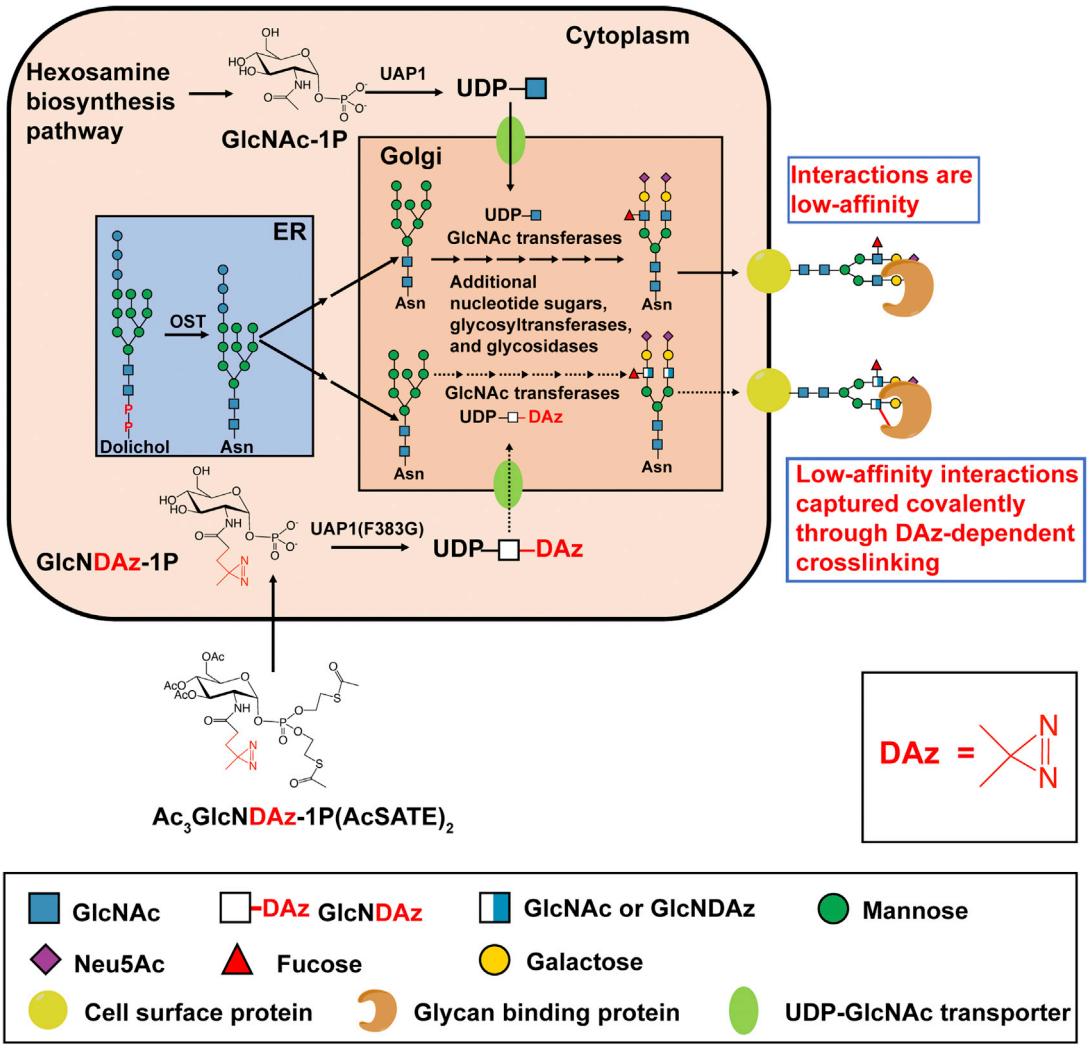


Figure 1. Strategy for metabolic production of photo-cross-linking N-glycans

N-glycans are synthesized in the ER and Golgi through the sequential actions of enzymes, including multiple GlcNAc transferases. N-linked glycoproteins are displayed on the cell surface, where glycans are recognized by extracellular GBPs. Individual binding interactions are typically low affinity, although high-avidity binding can be achieved through multivalency. To introduce diazirine, cells expressing UAP1(F383G) are cultured with Ac_3GlcNDAz-1P$(AcSATE)_2$. Ac_3GlcNDAz-1P$(AcSATE)_2$ is deprotected, potentially by intracellular esterases, and transformed into UDP-GlcNDAz. In this work, we asked whether UDP-GlcNDAz enters the Golgi and is a substrate for Golgi GlcNAc transferases. Cell-surface N-linked glycoproteins displaying the diazirine modification can be activated by UV light, resulting in covalent cross-linking to extracellular GBPs. OST, oligosaccharyltransferase.

See also Figure S1.

establishment of β1-2- and β1-6-antennae from α1-6-linked mannose in complex N-glycan biosynthesis. Several GlcNAc transferases involved in the biosynthesis of polyLacNAc, a terminal structure found on multiple types of glycans, and O-mannose glycans were also evaluated. While none of these enzymes appreciably transferred GlcNDAz, other functionally redundant GlcNAc transferases were not available. Therefore, it remains possible that GlcNDAz may also be incorporated into cell-surface O-linked glycans or polyLacNAc.

We compared the structures of MGAT1 and MGAT2 in their binding sites for UDP-GlcNAc donor. Because the structures of human MGAT1 and human MGAT2 with UDP-GlcNAc are not available, models were built using available structural data (Figure 2B) (Kadirvelraj et al., 2018; Unligil et al., 2000). For MGAT1, the N-acetyl group of UDP-GlcNAc is buried, with limited space to accommodate the diazirine, consistent with its inability to use UDP-GlcNDAz. In contrast, MGAT2 has a more surface-exposed donor-binding pocket, consistent with the modest activity with UDP-GlcNDAz we observed. Recent structural data for MGAT5 indicate that this enzyme also binds UDP-GlcNAc with the N-acetyl group exposed to solvent (Darby et al., 2020).

With the knowledge that UDP-GlcNDAz was produced and that some Golgi-resident GlcNAc transferases could accept it, we used glycoproteomics MS to determine whether GlcNDAz was incorporated into N-glycans. We employed a multistage enrichment strategy (Shajahan et al., 2020). Briefly, membrane proteins

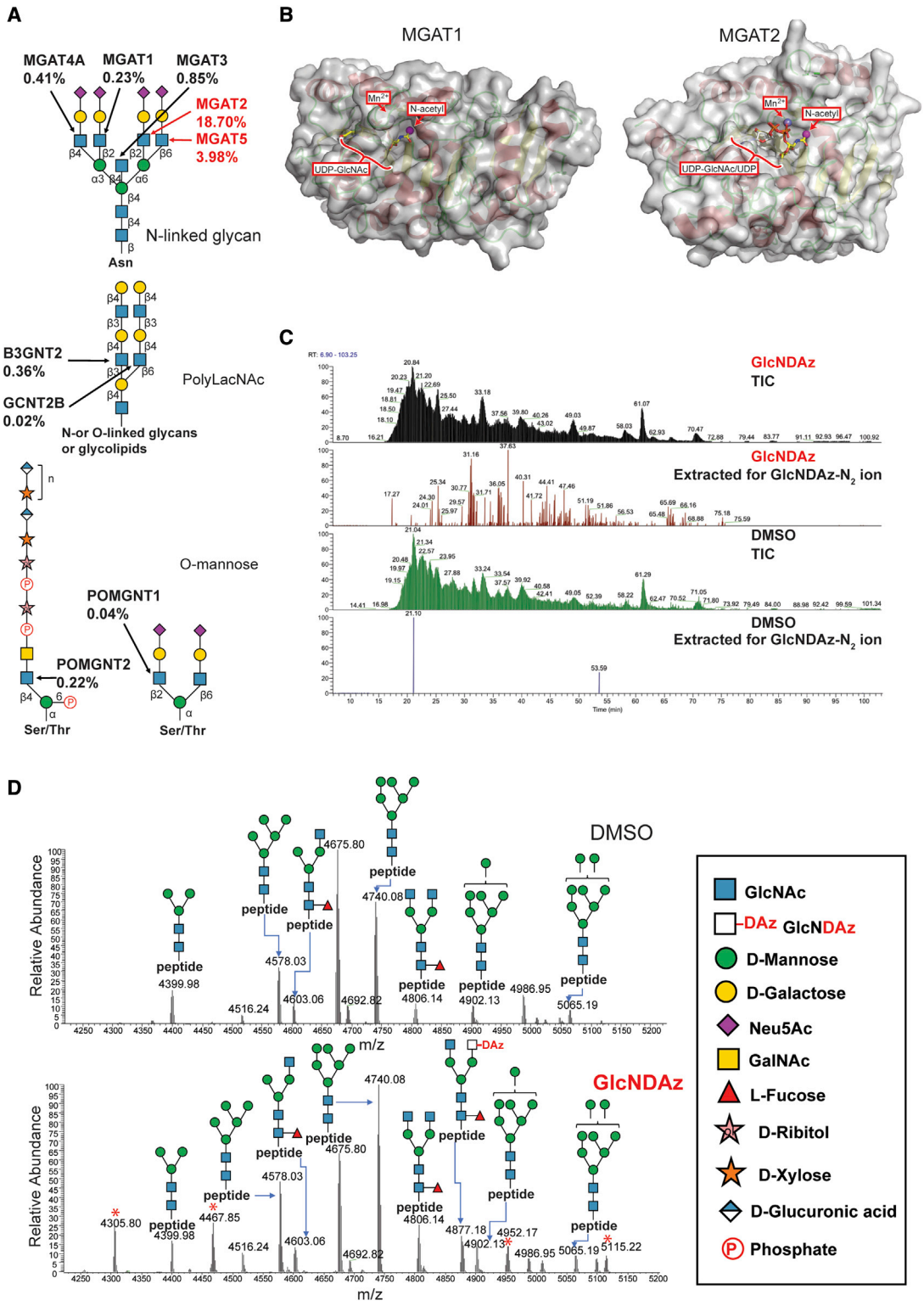


Figure 2. GlcNDAz is incorporated into N-glycans

(A) Biosynthetic roles of GlcNAc transferases were examined using recombinant Golgi GlcNAc transferases (Figure S2). Percentages represent relative activity of GlcNAc transferases in using UDP-GlcNDAz as a substrate compared with UDP-GlcNAc. Percentages are averaged from two trials.

(B) Modeled structures of human MGAT1 and MGAT2 (PDB: 5VCM) (Kadirvelraj et al., 2018) in complex with UDP-GlcNAc. The human MGAT1 structure was modeled based on the structure of the rabbit MGAT1:UDP-GlcNAc:Mn^{2+} (PDB: 1FOA) (Unligil et al., 2000). A magenta sphere marks the methyl group to which the diazirine is attached in GlcNDAz.

(legend continued on next page)

from T84 UAP1(F383G) cells cultured with Ac_3GlcNDAz-1P $(AcSATE)_2$ were enriched through two-step ultracentrifugation by differential dissolution of proteins (Nielsen et al., 2005). The membrane protein pellet was dissolved in urea lysis buffer with reducing agent, alkylated, and precipitated. Precipitated proteins were proteolyzed by trypsin, enriched by zwitterionic hydrophilic interaction liquid chromatography, and analyzed by liquid chromatography-tandem MS (LC-MS/MS) with higher-energy collisional dissociation (HCD) and product-triggered collision-induced dissociation (CID). When the total ion chromatogram (TIC) was extracted for the oxonium ions corresponding to GlcNDAz-N_2, a number of peaks were detected, which were absent from a sample of cells cultured without Ac_3GlcNDAz-1P$(AcSATE)_2$ (Figure 2C). Because previous studies showed that azide-functionalized GlcNAc (GlcNAz) can be epimerized to ManNAz by GlcNAc 2-epimerase (Luchansky et al., 2003), resulting in the azide group being incorporated into sialic acid (Shajahan et al., 2020), we asked whether GlcNDAz was converted to diazirine-modified sialic acid (SiaDAz) in a similar manner. Extraction for the SiaDAz-N_2 ion did not support the presence of SiaDAz (Figure S3A). MS/MS analysis was performed to gain insight into the position(s) at which GlcNDAz was incorporated. MS^2 fragmentation by HCD and CID on N-glycopeptides with GlcNDAz modification confirmed that GlcNDAz is on the terminal arms of the glycans but not in the glycan core. The presence of peaks corresponding to peptide with GlcNAc fragments (Y1 fragment) and lack of peaks corresponding to peptide with GlcNDAz fragments in HCD spectra indicate that the location of GlcNDAz is not at the glycan core. Moreover, CID fragmentation showed neutral loss of GlcNDAz fragments from glycan termini (Data S1). GlcNDAz was detected in biantennary and triantennary complex N-glycans of multiple proteins (Figure 3), and the extent of GlcNDAz incorporation varied among glycoforms examined (Figure S4). Also, the ratio of Hex:HexNAc found in some detected glycans implied that incorporated GlcNDAz could be further extended with Gal (Figure 3).

GlcNDAz engineering results in small perturbations to N-glycan structure

We considered whether GlcNDAz engineering perturbed N-glycan structures. Glycoproteomics analysis of an N-linked glycopeptide ^{38}VVRPDSELGERPPED**N**QSFQYDHEAFLGK^{66} from human reticulocalbin-1 revealed GlcNDAz incorporation on a biantennary glycoform. Comparison of data from control cells and Ac_3GlcNDAz-1P$(AcSATE)_2$-treated cells revealed only modest differences in the glycosylation pattern except for the presence of GlcNDAz-modified glycoforms in the treated case (Figure 2D). Thus, these data do not show evidence for dramatic alterations in N-glycan structure due to GlcNDAz engineering.

We also measured the effects of GlcNDAz incorporation on binding of lectins that recognize various glycan structures (Figure 4A) (Gao et al., 2019). T84 UAP1(F383G) cells were cultured with Ac_3GlcNDAz-1P$(AcSATE)_2$ or vehicle control. First, cells were fixed with paraformaldehyde and the binding of lectins to cell-surface glycoconjugates was measured by enzyme-linked lectin binding assay (ELLA), using multiple lectin concentrations (Figure S5). Statistical comparison was performed at non-saturating lectin concentrations (Figure 4B). One GlcNAc-recognizing lectin (succinylated wheat germ agglutinin, WGA) (Monsigny et al., 1980) showed no Ac_3GlcNDAz-1P$(AcSATE)_2$-dependent changes in lectin binding. However, Ac_3GlcNDAz-1P$(AcSATE)_2$ treatment led to an increase in the binding of another GlcNAc-recognizing lectin, *Griffonia* (*Bandeiraea*) *simplicifolia* lectin II (GSL II), that preferentially recognizes highly branched N-glycans (Nakamura-Tsuruta et al., 2006). Binding of *Maackia amurensis* lectin II (MAL II), which recognizes α2-3-linked sialic acid (Geisler and Jarvis, 2011), was unaffected by Ac_3GlcNDAz-1P$(AcSATE)_2$ treatment, but a small increase in binding of *Sambucus nigra* lectin (SNA), which recognizes α2-6-linked sialic acid, was observed. Binding of phytohemagglutinin-L (PHA-L), which recognizes β1-6 branches on N-glycans; of CTB, which recognizes multiple structures, including fucosylated glycans (Prudden et al., 2017); and of *Datura stramonium* lectin (DSL), which recognizes both polyLacNAc and β1-4 GlcNAc on α1-3 mannose (Cummings and Kornfeld, 1984), was also unaffected by Ac_3GlcNDAz-1P$(AcSATE)_2$ treatment. A lectin that recognizes polyLacNAc and β-linked galactose (human galectin-1) (Merkle and Cummings, 1987) (Nielsen et al., 2018; Stowell et al., 2008a), and a lectin that recognizes β1-4-linked galactose (*Erythrina crista-galli* lectin, ECA) also showed no change of binding with Ac_3GlcNDAz-1P$(AcSATE)_2$ treatment, although one polyLacNAc recognizing lectin (*Lycopersicon esculentum* lectin, LEL) showed an insignificant decrease in binding to cells cultured with Ac_3GlcNDAz-1P$(AcSATE)_2$. Next, we performed lectin blotting of lysates from cells treated with Ac_3GlcNDAz-1P$(AcSATE)_2$ or vehicle control (Figure 4C). Lectin blot analysis provides information about both intracellular and extracellular glycoproteins. Lectin blots confirmed the Ac_3GlcNDAz-1P$(AcSATE)_2$-dependent increase in GSL II binding, although the GSL II lectin blot could also reflect recognition of O-GlcNAc glycans (Ma and Hart, 2014). Also, a dramatic Ac_3GlcNDAz-1P$(AcSATE)_2$-dependent reduction in PHA-L binding was observed, mostly confined to a single band. Ac_3GlcNDAz-1P$(AcSATE)_2$-dependent increase in mobility of lectin-reactive species was observed for some lectins, notably DSL and LEL.

Taken together, GlcNAc-transferase assays, glycoproteomics analysis, and lectin binding data demonstrate that GlcNDAz can be metabolically incorporated into antennae of complex N-glycans. While diazirine-containing glycans are similar to those found in non-engineered cells, the process does result in some perturbations to N-glycan structure. Notably, increased GSL II binding was detected by both ELLA and lectin blot, suggesting that Gal may be added less efficiently to GlcNDAz than

(C) Glycopeptides from T84 UAP1(F383G) cells treated with Ac_3GlcNDAz-1P$(AcSATE)_2$ or DMSO were analyzed by mass spectrometry, and ion chromatograms were extracted for the GlcNDAz-N_2 ion. Chromatograms are from a single trial.

(D) Comparison of glycoforms detected on peptide ^{38}VVRPDSELGERPPED**NQS**FQYDHEAFLGK^{66} from human reticulocalbin-1 from T84 UAP1(F383G) cells treated with Ac_3GlcNDAz-1P$(AcSATE)_2$ or vehicle. Red asterisks label peaks (*m/z* 4,305.80, 4,467.85, 4,952.17, 5,115.22) assigned to other co-eluting glycopeptides. Spectrum is from a single trial.

See also Figures S2, S3, and S4 and Data S1.

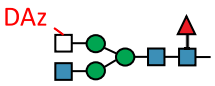
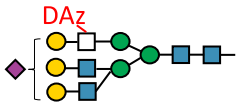
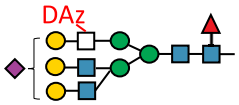
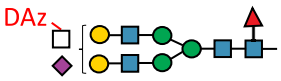
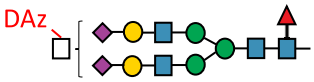
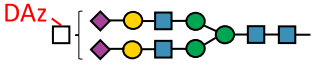
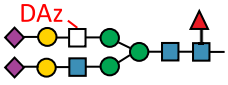
Glycan composition	Possible structure	Identified Protein (position of modification)
HexNAc(3)Hex(3)Fuc(1) HexNDAz(1)		RCN1(53), CELR3(1649)
HexNAc(4)Hex(6)NeuAc(1) HexNDAz(1)		LAMP2(49), CEAM5(375), LRP5(940), LAMA(1555)
HexNAc(4)Hex(6)Fuc(1)NeuAc(1) HexNDAz(1)		LAMP2(49, 257, 356), CEAM5(204), SCF(145), LAMP1(261), CEAM7(57), B3GL1(212), C4BPB(64), MPRD(83), AT1B1(158), FCGR1(240), RNF13(88), LIFR(787) , LY75(529)
HexNAc(4)Hex(5)Fuc(1)NeuAc(1) HexNDAz(1)		LAMP2(257), TSN8(118), CATL1(221), EDEM2(289), PTPRE(191), LAMP1(261), PACR(60), AGRG7(191), GRIK1(379)
HexNAc(4)Hex(5)Fuc(1)NeuAc(2) HexNDAz(1)		CEAM5(152, 204), TSN8(118), CATL1(221), NID2(693), IL6RB(553), LAMC1(1205), TTYH3(126)
HexNAc(4)Hex(5)NeuAc(2) HexNDAz(1)		ERAP2(294), CATL1(221), PKHD1(1377), SAMP(51), HPLN4(132), H6ST2 (404)
HexNAc(3)Hex(5)Fuc(1)NeuAc(2) HexNDAz(1)		I22R1(87), MA2B1(832), PAG15(273), OTOGL(2142)

Figure 3. GlcNDAz-containing glycopeptides

Glycopeptides extracted from T84 UAP1(F383G) cells treated with Ac_3GlcNDAz-1P(AcSATE)$_2$ were identified by mass spectrometry. Possible glycan structures are proposed based on glycan composition, MS^2 analysis, and biosynthesis pathways. We make the assumption that GlcNDAz is most likely to be found attached to α1-6Man in the N-glycan core, but data do not exclude other possible antennary arrangements. Data are from a single trial.

to GlcNAc residues. Similarly, the increased mobility of DSL and LEL reactive bands in the lectin blots suggests that GlcNDAz incorporation may interfere with antenna extension. Also, decreased PHA-L binding observed in the lectin blot but not the ELLA could reflect decreased N-glycan branching, particularly on intracellular glycoproteins. Nonetheless, glycoproteomics analysis demonstrates that Gal addition to GlcNDAz can occur, and mature GlcNDAz-containing glycans were detected, implying that GlcNDAz does not always act as a chain terminator. This observation is consistent with structural analysis of B4GALT1, a galactosyltransferase that is ubiquitously expressed and a likely candidate for catalyzing the addition of Gal to GlcNDAz (Hennet, 2002). Inspection of B4GALT1 crystal structures reveals that the N-acetyl group of GlcNAc on trisaccharide acceptors is localized in a relatively exposed binding pocket (Figure 4D) (Ramakrishnan et al., 2012; Ramasamy et al., 2005). In sum, GlcNDAz incorporation yields N-glycans that are similar but not identical to those produced by normal cells.

GlcNDAz is used to covalently capture glycan-mediated binding interactions

Having established the production of mature GlcNDAz-containing glycans, we tested whether metabolically incorporated diazirine could be used to photochemically cross-link known glycan recognition events. We selected two GBPs—galectin-1 and CTB—for which information about glycan specificity has been reported (Li et al., 2019; Prudden et al., 2017; Stowell et al., 2008a). T84 UAP1(F383G) cells were cultured with Ac_3GlcNDAz-1P(AcSATE)$_2$, then the GBP was added extracellularly and allowed to bind to cell-surface glycoconjugates at 4°C to prevent endocytosis (Figure 5A). Cells were irradiated to activate the cross-linker, after which they were immediately lysed. Lysates were analyzed by immunoblot using antibodies against

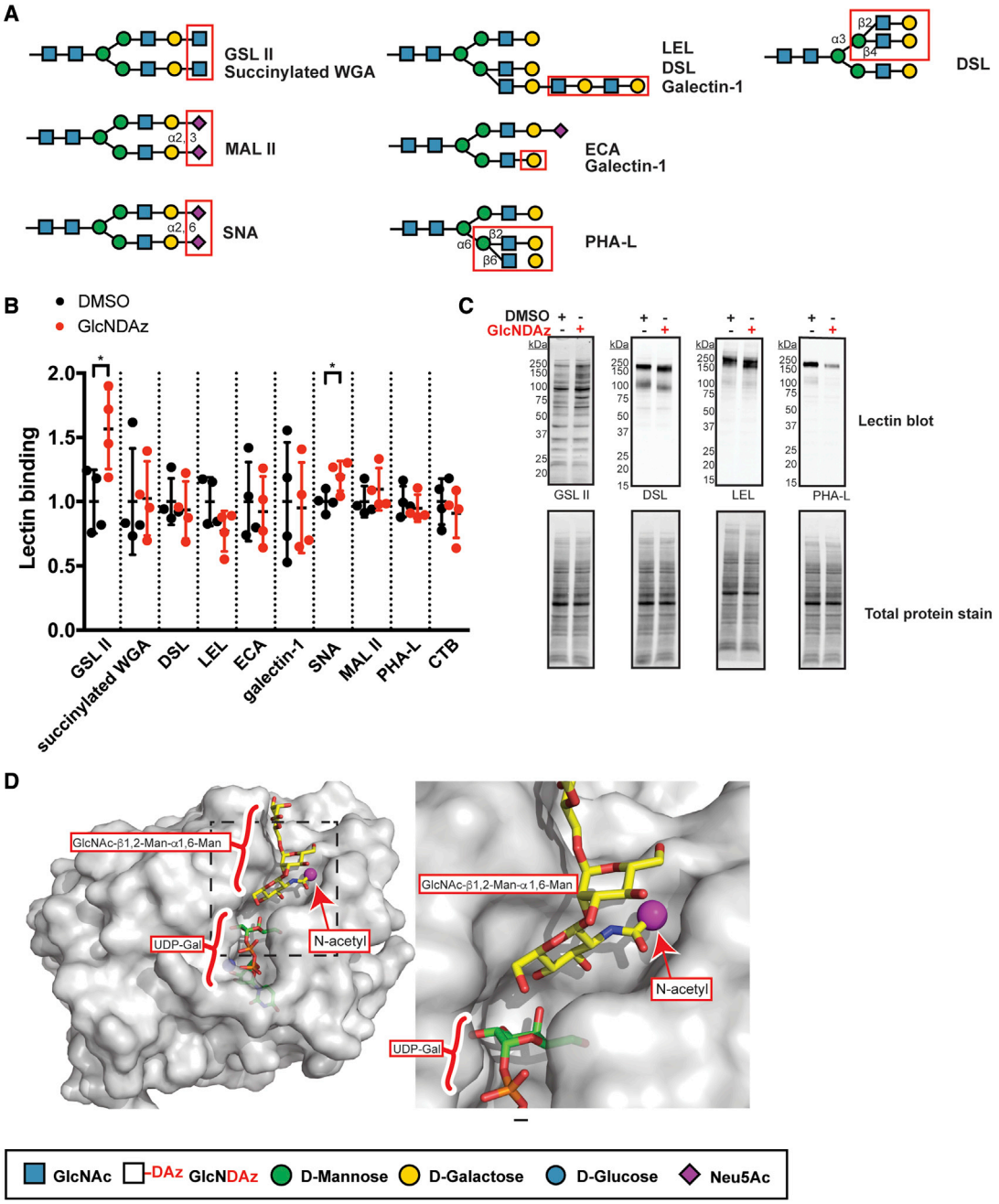


Figure 4. Effects of GlcNDAz incorporation on N-glycan structure

(A) Specificity of lectins used in (B) and (C).

(B) Lectin binding to T84 UAP1(F383G) cells treated with Ac_3GlcNDAz-1P(AcSATE)$_2$ at non-saturating concentrations (Figure S3) and normalized to vehicle-treated cells. Lectin concentrations used: GSL II, 1 μg/mL; succinylated WGA, 0.1 μg/mL; DSL, 1 μg/mL; LEL, 1 μg/mL; ECA, 0.1 μg/mL; galectin-1, 1 μg/mL; SNA, 3 μg/mL; MAL II, 3 μg/mL; PHA-L, 3 μg/mL; CTB, 3 μg/mL. Data represent the mean ± SD from four biological replicates, each of which is averaged from three technical replicates. p Values were determined using two-tailed unpaired t test. A single asterisk (*) indicates a p value no larger than 0.05.

(C) Lectin blots of total lysates from T84 UAP1(F383G) cells treated with Ac_3GlcNDAz-1P(AcSATE)$_2$ or vehicle. Blots are representative of biological triplicates.

(D) Structure of human B4GALT1 (overlay of PDB: 2AEC [Ramasamy et al., 2005] and PDB: 1TVY [Ramakrishnan et al., 2006]) in complex with UDP-Gal, and the trisaccharide acceptor-containing terminal GlcNAc is shown with zoom-in for the area indicated by the dotted box. A magenta sphere marks the methyl group to which the diazirine is attached in GlcNDAz.

See also Figure S5.

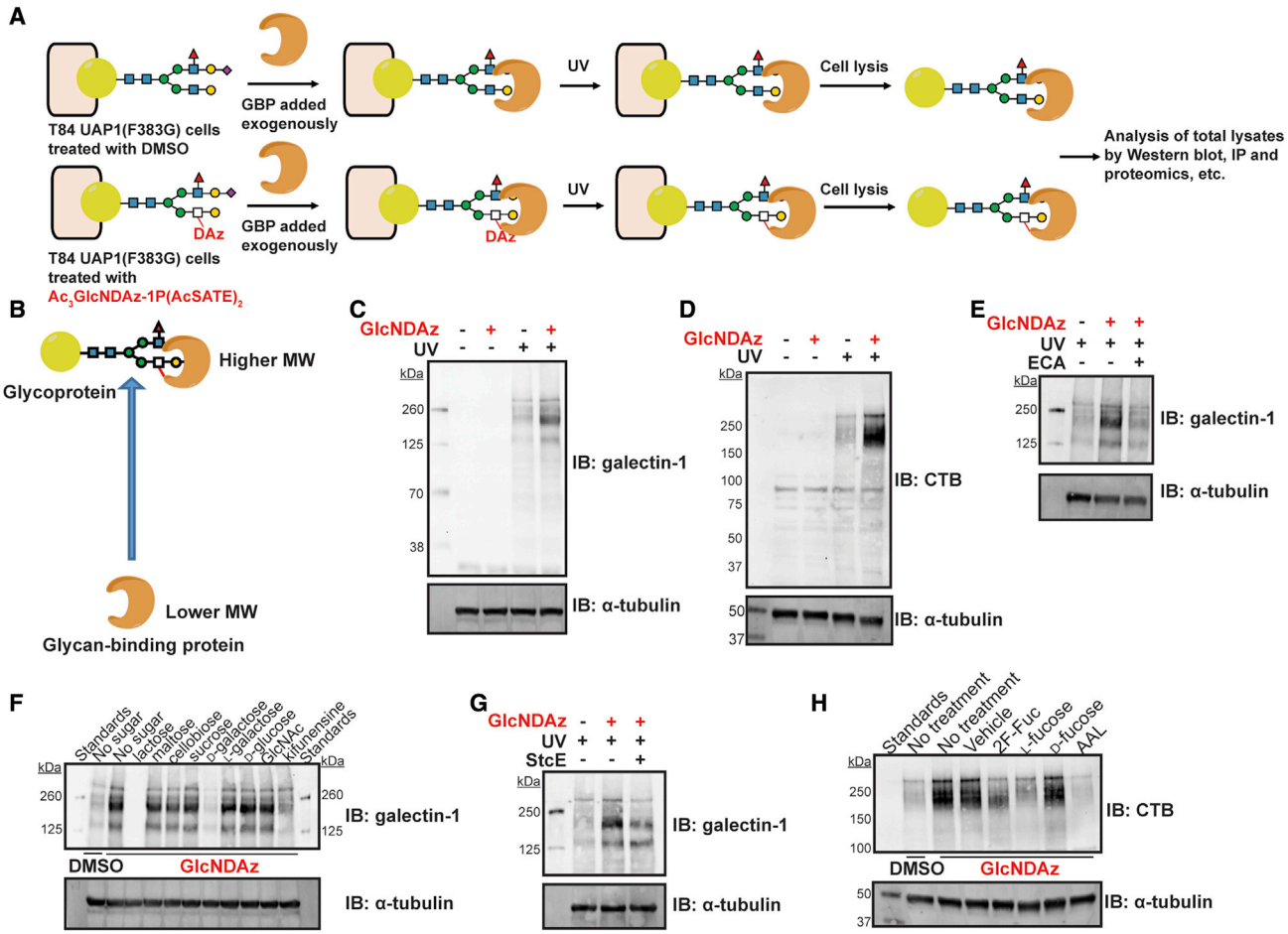


Figure 5. GlcNDAz can be used to covalently capture glycan-mediated binding interactions

(A) Scheme of cross-linking experiment.

(B) Scheme of detecting cross-linking complexes by immunoblot analysis of total cell lysates. Cross-linking of human galectin-1 (~15 kDa) and CTB (~14 kDa) to larger cell surface glycoproteins results in a shift of galectin-1 and CTB to apparent higher molecular weight.

(C) Cross-linking of galectin-1 to glycoproteins from T84 UAP1(F383G) cells. IB, immunoblot.

(D) Cross-linking of cholera toxin to glycoproteins from T84 UAP1(F383G) cells.

(E) Cross-linking of galectin-1 with pre-treatment with 40 μg/mL ECA lectin.

(F) Cross-linking of galectin-1 in the presence of 100 mM mono- or disaccharides, or with pre-treatment of cells with 1 μg/mL kifunensine for 72 h.

(G) Cross-linking of galectin-1 to glycoproteins with pre-treatment by StcE.

(H) Cross-linking of CTB with cells cultured with 200 μM 2F-fucose for 72 h, pre-treatment with 40 μg/mL lectin AAL, or in the presence of 100 mM L-fucose or D-fucose. All blots are representative of biological triplicates.

GBPs. Both GBPs have low molecular weights (~15 kDa for galectin-1 and ~12 kDa for CTB) and are not retained in low-percentage gels. However, when GBPs are cross-linked to N-linked glycoproteins, their apparent molecular weight increases and cross-linked species can be detected (Figure 5B). For both galectin-1 (Figure 5C) and CTB (Figure 5D), no cross-linked species were observed when UV was not applied, and only minimal cross-linking was detected when Ac_3GlcNDAz-1P$(AcSATE)_2$ was omitted, while robust cross-linking was detected when Ac_3GlcNDAz-1P$(AcSATE)_2$ was included and UV was applied. Background cross-linking observed in the absence of the cross-linker might be due to radical species formed by UV irradiation of aromatic amino acid residues (Russ et al., 2005).

We interrogated the nature of the galectin-1-glycoprotein cross-linked species. Galectin-1 preferentially binds Galβ1-4GlcNAc (N-acetyllactosamine, LacNAc), and inclusion of the ECA lectin, which also recognizes LacNAc, resulted in reduced cross-linking (Figure 5E). Galectin-1 binding to LacNAc can be competitively inhibited by Galβ1-4Glc (lactose, Lac), so we also tested whether different disaccharides and monosaccharides interfered with cross-linking (Lantéri et al., 2003; Shimura et al., 2002). Cross-linking was eliminated by inclusion of 100 mM Lac, but not by Glcα1-6Glc (maltose), Glcβ1-4Glc (cellobiose), or Glcα1-2Fruβ (sucrose) (Figure 5F). Similarly, the terminal sugar of LacNAc, D-Gal, inhibited galectin-1-glycoprotein cross-linking, but L-Gal, D-Glc, and D-GlcNAc did not. Also, culturing cells with kifunensine, an inhibitor of N-glycan

maturation (Elbein et al., 1990), reduced cross-linking, indicating that at least some glycans recognized by galectin-1 are displayed on N-linked glycoproteins (Figure 5G). However, pretreatment of cells with the protease StcE, which cleaves adjacent to GalNAc-type O-linked glycosylation sites (Malaker et al., 2019), also reduced cross-linking. This result suggests that some cross-linking may occur through recognition of O-linked glycans. In sum, the galectin-1 cross-linking results are consistent with previous studies demonstrating that galectin-1 binds to cells through both N-linked and O-linked glycans (Nielsen et al., 2018; Seelenmeyer et al., 2003).

We performed similar analysis for CTB (Figure 5H). CTB recognizes multiple glycans, including the canonical receptor ganglioside GM1, as well as fucosylated glycoconjugates, which mediate a large fraction of CTB binding to T84 cells (Wands et al., 2015). As predicted, culturing cells with a metabolic inhibitor of fucosylation, 2F-fucose (Rillahan et al., 2012), reduced cross-linking. Inclusion of L-Fuc, but not D-Fuc, in the binding buffer reduced cross-linking. Finally, inclusion of *Aleuria aurantia* lectin (AAL), which competes with CTB for binding to fucosylated glycoproteins (Wands et al., 2015), resulted in reduced cross-linking.

Together, these results demonstrate that metabolically incorporated GlcNDAz can be used to covalently cross-link known GBP-glycoprotein binding interactions, and that the extent of cross-linking observed depends on the presence and accessibility of preferred glycan ligands for the GBP.

GlcNDAz-enabled proteomics facilitates identification of galectin-1 binding partners

Covalent cross-linking offers the capacity to isolate and identify glycoprotein binding partners of GBPs. To do so, we cultured T84 UAP1(F383G) cells with Ac_3GlcNDAz-1P$(AcSATE)_2$ or vehicle and then added biotinylated galectin-1 extracellularly and irradiated with UV. Cross-linked complexes were purified using streptavidin beads. Purified material was separated into three regions of an SDS polyacrylamide gel, digested with trypsin, and analyzed by LC-MS/MS. Protein abundance was evaluated and compared with control samples from which Ac_3GlcNDAz-1P$(AcSATE)_2$, UV irradiation, and biotinylated galectin-1 were omitted (Table S1). Proteins were considered hits if they were enriched more than 5-fold in the experimental sample, if the abundance was greater than 100,000, and with a minimum number of peptide spectrum matches of 2. Proteins that were identified but not quantified and potential contaminants were excluded. The high-molecular-weight region of the gel yielded 86 hits, the medial-molecular-weight region yielded 73 hits, and the low-molecular-weight region yielded 30 hits (Figure 6A). A total of 103 protein hits were identified (Table S1). More than half of the hits were identified in both biological replicates (Figure 6B). Candidate galectin-1 binding partners were further analyzed with DAVID Informatics Resource 6.8 (Huang et al., 2009a, 2009b). Among the 102 recognized hits (Table S1), the majority are known to be glycosylated: 70 proteins were annotated glycoproteins based on UniProt keywords and 67 proteins were annotated to have N-linked glycosylation sites based on UniProt. We also used the GlyGen Bioinformatics Resource, which catalogs experimentally demonstrated glycosylation sites (York et al., 2020). Narrowing analysis to the 46 hits detected in the experimental sample but not in the control sample (i.e., those displaying infinite enrichment) in at least one region of the gel, we found that GlyGen cites experimental evidence of N-glycans for 40 hits, while additional literature reports evidence for N-linked glycosylation for at least three additional hits (Kariya et al., 2008; Nittis and Gitlin, 2004: Samanta et al., 2011). Thus, from the most highly enriched subset of 46 proteins, 93% are known N-glycoproteins. We conclude that GlcNDAz cross-linking and enrichment is effective at identifying glycoproteins whose characteristics are consistent with galectin-1 binding.

We evaluated whether previously reported galectin-1 binding partners were identified. Carcinoembryonic antigen (CEA), a galectin-1 ligand identified in an early study of the colon carcinoma cell line KM12, was among the top hits from all three cross-linked regions (Ohannesian et al., 1994). Galectin-1 ligands reported from other cell lines, such as CDCP1, laminin, integrin β1, and CD44 (Camby et al., 2006; Elola et al., 2005; Ito and Ralph, 2012; Law et al., 2016) (Moiseeva et al., 2003), were found among the hits (Table S1). Cross-linking of CEA, CDCP1, and LY75 to galectin-1 was further confirmed by immunoprecipitation (IP) (Figures 6C–6E and S6) and immunoblot, where a band(s) migrating near or slightly above the molecular weight of the putative binding partner can be detected by anti-galectin-1 antibody. This result demonstrates direct cross-linking of galectin-1 to each of these binding partners.

LAMP1 and LAMP2, identified as galectin-1 ligands from KM12 cells and other cell lines (Do et al., 1990; Obermann et al., 2017; Ohannesian et al., 1994), were not enriched in the proteomics analysis. We also failed to detect cross-linked complexes between LAMP1 and galectin-1 in IP-enriched samples (Figure 7A), even though LAMP1 is readily detected in these cells (Figures S7A and S7B). We speculated that our inability to detect galectin-1 cross-linking to LAMP1 could be because the extracellularly added galectin-1 was unable to contact LAMP1 localized to the lysosome or other intracellular compartments. Indeed, galectin-1 displays a different cross-linking pattern when cross-linking was performed in lysates rather than intact cells (Figure 7B, input lanes). Further, when cross-linking was performed in lysates, cross-linked complexes between LAMP1 and galectin-1 were enriched by LAMP1 IP and detected by galectin-1 immunoblot (Figure 7B). In contrast, CEA, which was highly enriched in proteomics analysis, was cross-linked to galectin-1 both in lysates and in intact cells (Figures S7C and S7D).

DISCUSSION

MGE is a powerful method to introduce unnatural functional groups into cellular glycans (Du et al., 2009). The relaxed substrate tolerance of metabolic enzymes is critical for MGE, but can also allow metabolic interconversions that yield unnatural sugar incorporation at unexpected or undesired locations. For example, MGE conducted with GlcNAz has revealed metabolic conversion to azide-modified N-acetylneuraminic acid (Saxon et al., 2002; Shajahan et al., 2020). Here we used MGE to introduce diazirine onto GlcNAc residues of N-glycans, which were characterized by MS-based glycoproteomics. Diazirine was found attached to antennary GlcNAc, and no SiaDAz was detected. One possible explanation for different outcomes for

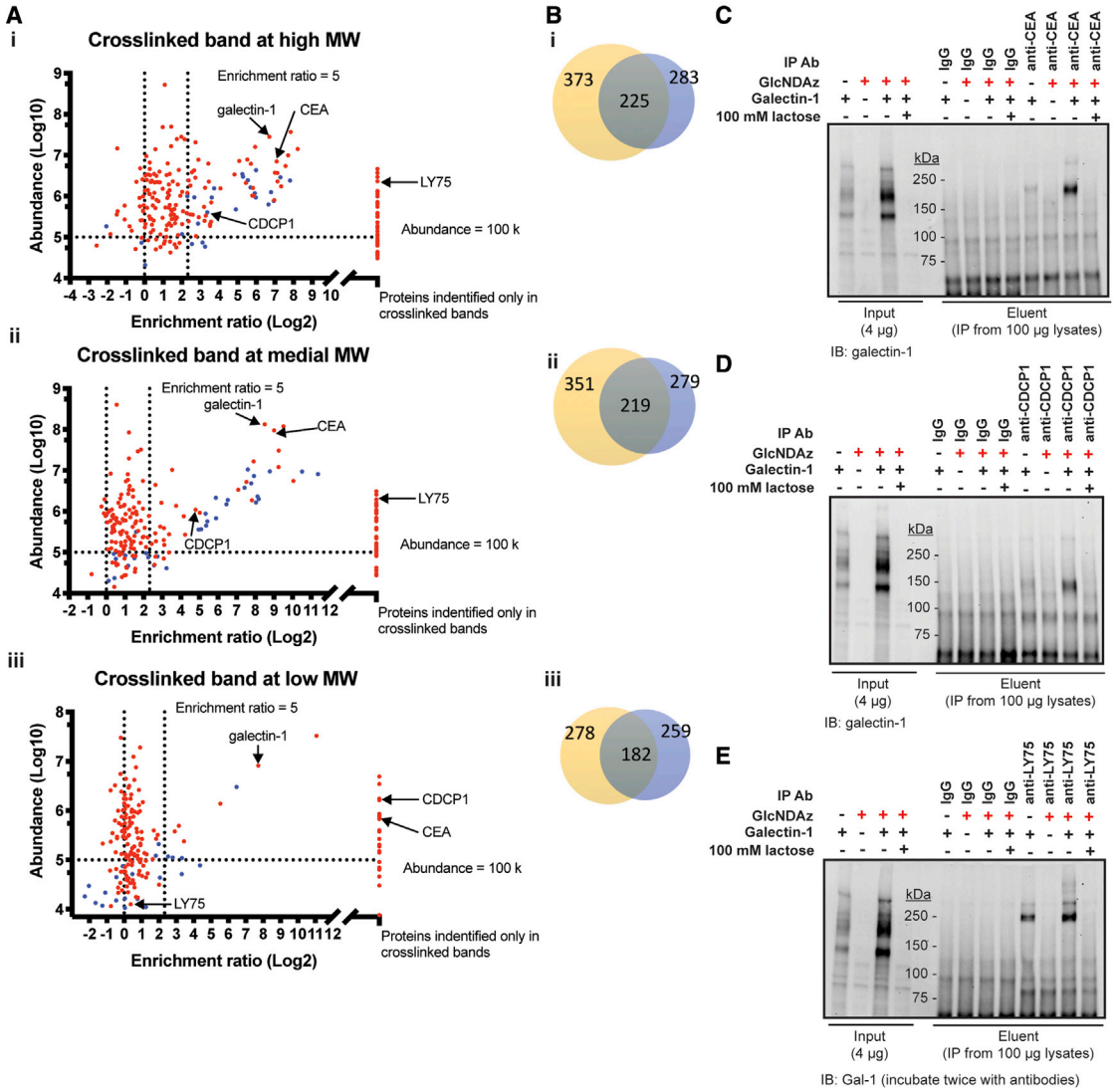


Figure 6. GlcNDAz-enabled proteomics facilitates identification of candidate galectin-1 binding partners

(A) Scatterplots of proteins from T84 UAP1(F383G) cells identified in the three major cross-linked bands in two biological replicates. Abundance and fold-enrichment represent the average of two biological replicates. Proteins detected in cross-linked but not control samples are shown on the right of each plot. Proteins detected in cross-linked samples in two replicates but detected in control samples in only one replicate are represented with blue symbols. For these proteins, fold-enrichment from the trial in which proteins were identified in both cross-linked and control samples was used in the plot.

(B) Venn diagram of proteins identified from each region in two replicates.

(C–E) CEA (C), CDCP1 (D), and LY75 (E) were enriched by immunoprecipitation with anti-CEA, anti-CDCP1, and anti-LY75 antibodies and cross-linked complexes were detected with anti-galectin-1 antibody by immunoblot. Confirmation of successful immunoprecipitation is presented in Figure S6. See also Table S1. (C–E) are representative of biological duplicates using 100 μg/mL galectin-1 for cross-linking. In a third trial conducted with 30 μg/mL galectin-1, similar results, albeit with fainter cross-linked bands, were obtained.

azide-modified versus diazirine-modified GlcNAc is the size of the unnatural functional groups. Indeed, RENBP, which interconverts GlcNAc and ManNAc, epimerizes azide-modified ManNAc 345 times more rapidly than a ManNAc analog with a levulinoyl modification that is about the same size as the diazirine modification employed here (Luchansky et al., 2003). Alternatively, our strategy of delivering a GlcNAc-1-P analog and overexpressing a UAP1 mutant with relaxed specificity (Yu et al., 2012) may direct more of the unnatural sugar flux through UDP-GlcNAc analogs, rather than toward ManNAc and sialic acid analogs. Also, we found that the Golgi-resident GlcNAc transferase MGAT2 uses the UDP-GlcNDAz, although with reduced efficiency compared with the natural substrate. This preference is similar that of OGT, which displayed a 12-fold difference in catalytic efficiency between UDP-GlcNAc and UDP-GlcNDAz (Rodriguez et al., 2015). We propose that GlcNDAz is most likely to be found attached to α1-6Man in the N-glycan core; isolation of GlcNDAz-modified N-glycans or N-glycopeptides and analysis by NMR would be required to demonstrate GlcNDAz positions more definitively. While the efficiency of GlcNDAz incorporation was

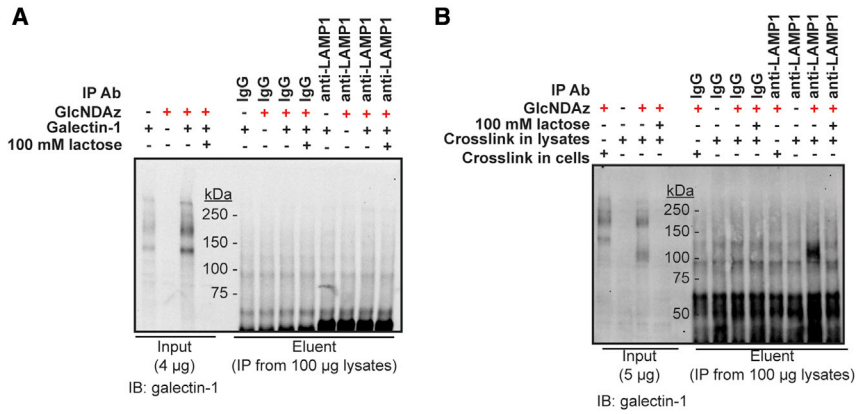


Figure 7. GlcNDAz-mediated cross-linking of LAMP1 to galectin-1 occurs in cell lysates but not intact cells

(A) T84 UAP1(F383G) cells were cultured with $Ac_3GlcNDAz\text{-}1P(AcSATE)_2$ or vehicle. Galectin-1 was added to a final concentration of 100 µg/mL, then the samples were UV irradiated. Cells were lysed and immunopurified for LAMP1. Immunoblot analysis was performed with a galectin-1 antibody. Data are representative of biological duplicates using 100 µg/mL galectin-1 for cross-linking. In a third trial conducted with 30 µg/mL galectin-1, similar results were obtained.

(B) T84 UAP1(F383G) cells were cultured with $Ac_3GlcNDAz\text{-}1P(AcSATE)_2$ or vehicle. Cells were lysed, then 30 µg/mL galectin-1 was added and UV irradiation applied. Immunopurification was performed with a LAMP1 antibody. Immunoblot analysis was performed with a galectin-1 antibody. Data are representative of biological triplicates.

See also Figure S7.

sufficient to support cross-linking, further improvements may be possible through targeted mutations to MGAT2 or other enzymes.

Binding of galectin-1 and CTB to T84 cell surfaces was unaffected by $Ac_3GlcNDAz\text{-}1P(AcSATE)_2$ treatment, suggesting that the MGE did not eliminate production of preferred glycan ligands for these GBPs. Nonetheless, knowledge of even modest perturbations to glycan structures is essential for reliable use of this tool. While MS-based glycoproteomics demonstrated that mature GlcNDAz-containing N-glycans were produced, lectin-binding experiments revealed some alterations in glycan structures. One possible explanation is that B4GALTs may inefficiently extend GlcNDAz-containing glycans, resulting in fewer mature terminal structures and shorter polyLacNAc chains. Also, the MGE approach results in a high concentration of UDP-GlcNDAz. This metabolic perturbation could affect the production of GlcNAc/GlcNDAz-containing glycans. Elevated UDP-GlcNAc also has the potential to lead to increased sialylation through activation of GNE (Hinderlich et al., 2015), consistent with the small increase in SNA binding.

GlcNDAz cross-linking can offer insight into the preferred glycan ligands for a GBP. For example, the intensity of CTB-glycoprotein cross-linking was reduced when preferred fucosylated ligands were absent. Also, glycoproteins that display the preferred ligands for a GBP can be identified by covalent cross-linking, followed by purification and proteomics analysis. Known galectin-1 binding partners, such as CEA (Ohannesian et al., 1994), CDCP1 (Law et al., 2016), and integrin β4 (Moiseeva et al., 2003), were identified through such a workflow, which also revealed additional potential galectin-1 partners. The common occurrence of N-linked glycoproteins among the most highly enriched hits implies a high specificity for identification of genuine galectin-1 binding partners, but further improvements may be possible. These might include the use of PNGase F to specifically elute cross-linked partners with N-glycans or the use of lactose-pre-treated cells as a comparator in the proteomics analysis. We confirmed cross-linking to galectin-1 for multiple proteins from the proteomics hit list, including one, LY75, that we believe was not previously identified as a galectin-1 binding partner. In this method, the appearance of a cross-linked complex is indicative of a direct interaction between the cross-linking partners. This represents an advantage over traditional immunopurification approaches in which a third party may mediate an observed binding interaction.

Another factor affecting GlcNDAz cross-linking is the incorporation efficiency of GlcNDAz at different glycan positions. GlcNDAz may not be incorporated into the preferred glycan structures of some GBPs. For example, glycoproteomic analysis did not detect GlcNDAz incorporation into the $Man_3GlcNAc_2$ core of N-glycans, which is recognized by some E3 ligases (Yoshida et al., 2019). Thus, our data confirm the incorporation of GlcNDAz into N-glycans and also set limits on the use of this tool. However, the decreased galectin-1 cross-linking to T84 UAP1(F383G) cells treated with StcE also suggests the possible incorporation of GlcNDAz into O-GalNAc glycans, which we plan to explore in future work. Expanding the positions of GlcNDAz incorporation by structure-based engineering of GlcNAc-transferases, such as MGAT1 and POMGNT1, may also facilitate broader use.

Proteomics analysis of galectin-1 cross-linking partners offers some indication of the types of biological insights that may be obtained with this method. Typically, discovery of GBP binding partners is performed by affinity purification from cell lysates, conditions that disrupt the normal cellular architecture and may allow interactions among proteins that normally reside in distinct compartments. Indeed, an early study employing immobilized galectin-1 and KM12 cell extracts identified CEA, LAMP1, and LAMP2 as galectin-1 binding partners (Ohannesian et al., 1994). The ability of LAMP1 and LAMP2 to interact with galectin-1 was confirmed in lysates from other cell types (Do et al., 1990; Obermann et al., 2017). However, our proteomics analysis of the binding partners of extracellularly added galectin-1 identified the CEA-displaying protein CEACAM5, but not LAMP1. Because glycoproteomic analysis confirmed the incorporation of GlcNDAz into glycans on multiple asparagine residues of LAMP1, we speculated that the absence of LAMP1 and LAMP2 in the proteomics analysis arose from the fact that the cross-linking was performed in intact cells, where extracellularly

added galectin-1 did not contact lysosomal proteins. Indeed, when cells were lysed to allow mixing of intracellular and extracellular components prior to UV irradiation, we observed cross-linking of LAMP1 to galectin-1. Thus, the ability to report on cellular context is an advantage of the cross-linking method. We plan to examine whether cytoplasmic galectin-1 is capable of cross-linking to lysosomal proteins such as LAMP1 and LAMP2, and whether the extent of these interactions changes under conditions that induce lysosomal damage pathways, which expose lysosomal glycoproteins to cytoplasmic GBPs and recruit multiple galectins, including galectin-1, to damaged lysosomes (Aits et al., 2015; Jia et al., 2018, 2020).

SIGNIFICANCE

N-linked glycoproteins are key components of many extracellular recognition events. Their N-glycans are recognized by a variety of glycan-binding proteins. These glycan recognition events regulate diverse cellular processes, including cell-cell interactions, signal transduction, and host-microbe recognition. However, glycan recognition events are often low affinity and transient, characteristics that pose challenges for identifying binding partners through traditional affinity purification approaches. Here we describe a metabolic labeling method to incorporate the diazirine photo-cross-linking functional group onto GlcNAc residues of N-glycans. Importantly, the method results in only modest perturbations to N-glycan structures. We demonstrate that the metabolically incorporated diazirine can be activated by UV light, yielding covalent cross-linking between N-linked glycoproteins and extracellular glycan-binding proteins. The resulting cross-linked complexes can be analyzed by biochemical methods including immunoblot, immunoprecipitation, and proteomics mass spectrometry analysis. Pharmacological agents that modulate glycan structure as well as protein and glycan reagents that competitively interfere with glycan binding can be used in conjunction with immunoblot analysis of cross-linked complexes, revealing information about glycan features that are required for specific glycan recognition events. Protein components of cross-linked complexes can be identified by proteomics mass spectrometry analysis of the cross-linked complexes. Finally, information about the cellular context of glycan recognition events can be discerned by evaluating the conditions under which cross-linking is observed. These results support the utility of photo-cross-linking cell-surface sugars in identifying and defining glycan-dependent binding interactions within their native context.

STAR★METHODS

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SUPPLEMENTAL INFORMATION

Supplemental information can be found online at https://doi.org/10.1016/j.chembiol.2021.07.007.

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AUTHOR CONTRIBUTIONS

Conceptualization, H.W. and J.J.K.; experimental investigation, H.W., A.S., J.-Y.Y., and E.C.; reagents, A.M.W., C.M.A., and S.R.S.; data curation, H.W. and A.S.; writing – original draft, H.W. and J.J.K.; writing – review & editing, H.W., A.S., J.-Y.Y., K.W.M., and J.J.K.; supervision, K.W.M., P.A., and J.J.K.; project administration, J.J.K.; funding acquisition, S.R.S., K.W.M., P.A., and J.J.K.

DECLARATION OF INTERESTS

K.W.M. acknowledges ownership interest and roles as president and CEO of Glyco Expression Technologies, Inc., a biotechnology spinout commercializing recombinant glycosyltransferases, and may conceivably profit from the results described herein. J.J.K. is a member of the advisory board for *Cell Chemical Biology*. Other authors declare no competing interests.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Rabbit monoclonal anti-galectin-1	Cell Signaling Technology (Danvers, MA, USA)	Cat# 12936; RRID: AB_2798065
Goat polyclonal anti-galectin-1	R&D Systems (Minneapolis, MN, USA)	Cat# AF1152; RRID: AB_2136626
Rabbit monoclonal anti-LAMP1	Cell Signaling Technology	Cat# 9091; RRID: AB_2687579
Rabbit monoclonal anti-CDCP1	Cell Signaling Technology	Cat# 13794; RRID: AB_2798314
Rabbit polyclonal anti-CDCP1	Cell Signaling Technology	Cat# 4115; RRID: AB_2078818
Rabbit monoclonal anti-LY75	Abcam (Cambridge, UK)	Cat# ab124897; RRID: AB_10976058
Rabbit polyclonal anti-cholera toxin B subunit	Abcam	Cat# ab34992; RRID: AB_726859
Mouse monoclonal anti-α-tubulin	Sigma (St. Louis, MO, USA)	Cat# T6199; RRID: AB_477583
Mouse monoclonal anti-CEA	Thermo Fisher Scientific (Waltham, MA, USA)	Cat# MA5-14675; RRID: AB_10985171
IRDye® 680 RD goat anti-mouse IgG (H + L)	LI-COR Biosciences (Lincoln, USA)	Cat# 926-68,070; RRID: AB_10956588
IRDye® 800 CW goat anti-mouse IgG (H + L)	LI-COR Biosciences	Cat# 926-32210; RRID: AB_621842
IRDye® 800 CW goat anti-rabbit IgG (H+L)	LI-COR Biosciences	Cat# 926-32211; RRID: AB_621843
IRDye® 800 CW donkey anti-goat IgG (H+L)	LI-COR Biosciences	Cat# 925-32214; RRID: AB_2687553
Chemicals, peptides, and recombinant proteins		
Ac_3GlcNDAz-1P(AcSATE)$_2$	(Yu et al., 2012) Wuxi AppTec (Tianjin)	N/A
UDP-GlcNDAz	(Yu et al., 2012) Wuxi AppTec (Tianjin)	N/A
UDP-GlcNAc	Promega (Madison, WI, USA)	Cat# V707A
Dimethyl sulfoxide	Fisher Scientific (Hampton, NH, USA)	Cat# BP231-100
Lactose	Sigma	Cat# 61339-25G
Cellobiose	Sigma	Cat# 22150-10G
D-galactose	Sigma	Cat# G0750-100G
L-galactose	Sigma	Cat# G7134-100MG
D-fucose	Sigma	Cat# F8150-1G
L-fucose	Sigma	Cat# F2252-5G
D-glucose	Sigma	Cat# G7021-100G
N-acetylglucosamine	Sigma	Cat# A3286-25G
Maltose	MP Biomedicals (Santa Ana, CA, USA)	Cat# 190237
Sucrose	Fisher Scientific	Cat# BP220-1
Kifunensine	Sigma	Cat# K1140-1MG
2F-peracetyl-fucose	Millipore (Burlington, MA, USA)	Cat# 344827-10MG
Biotinylated *Griffonia* (*Bandeiraea*) *Simplicifolia* Lectin II	Vector Laboratories (Burlingame, CA, USA)	Cat# B-1215
Biotinylated Succinylated Wheat Germ Agglutinin	Vector Laboratories	Cat# B-1025S
Biotinylated *Datura Stramonium* Lectin	Vector Laboratories	Cat# B-1185
Biotinylated *Lycopersicon Esculentum* (Tomato) Lectin	Vector Laboratories	Cat# B-1175
Biotinylated *Erythrina Cristagalli* Lectin	Vector Laboratories	Cat# B-1145
Biotinylated *Sambucus Nigra* Lectin	Vector Laboratories	Cat# B-1305
Biotinylated *Maackia Amurensis* Lectin II	Vector Laboratories	Cat# B-1265

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Biotinylated *Phaseolus Vulgaris* Leucoagglutinin	Vector Laboratories	Cat# B-1115
Aleuria Aurantia Lectin	Vector Laboratories	Cat# L-1390
Cholera toxin B subunit	Sigma	Cat# C9903
Recombinant cholera toxin B subunit, biotinylated	Thermo Fisher Scientific	Cat# C-34779
Recombinant human galectin-1	(Wu et al., 2021)	N/A
Recombinant human galectin-1, biotinylated	(Stowell et al., 2008a; Stowell et al., 2008b)	N/A
IRDye® 800 CW streptavidin	LI-COR Biosciences	Cat# 926-32230
Streptavidin-POD conjugate	Sigma	Cat# 11089153001
O-phenylenediamine dihydrate	Sigma	Cat# P1526-10G
Normal rabbit IgG	Millipore	Cat# NI01-100UG
Normal mouse IgG	Millipore	Cat# NI03-100UG
Carbo-free Blocking Solution	Vector Laboratories	Cat# SP-5040
Intercept™ (TBS) Blocking Buffer and Diluent Kit	LI-COR Biosciences	Cat# 927-66003
Dynabeads™ M-280 streptavidin	Thermo Fisher Scientific	Cat# 11206D
Dynabeads™Protein G	Thermo Fisher Scientific	Cat# 10004D
cOmplete™, Mini, EDTA-free Protease Inhibitor Cocktail	Sigma	Cat# 11836170001
20% Paraformaldehyde (formaldehyde) aqueous solution	Electron Microscopy Sciences (Hatfield, PA, USA)	Cat# 15713
StcE	(Malaker et al., 2019)	N/A
Man3GlcNAc2-Asn	(Kadirvelraj et al., 2020)	N/A
GlcNAcMan3GlcNAc2-Asn	(Kadirvelraj et al., 2020)	N/A
GlcNAc2Man3GlcNAc2-Asn	(Kadirvelraj et al., 2020)	N/A
O-Man glycopeptide 317	(Halmo et al., 2017)	N/A
Lacto-N-neotetraose	Carbosynth (Compton, UK)	Cat# OL05765
Critical commercial assays		
ProteoExtract® Glycopeptide Enrichment Kit	Millipore	Cat# 72103-3
UDP-Glo™ Glycosyltransferase Assay	Promega	Cat# V6961
Revert™ 700 Total Protein Stain Kit	LI-COR Biosciences	Cat# 926-11016 or 926-11010
Pierce™ BCA Protein Assay Kit	Thermo Fisher Scientific	Cat# 23225
Pierce™ Silver Stain for Mass Spectrometry	Thermo Fisher Scientific	Cat# 24600
Deposited data		
Glycoproteomics raw data and results	This paper	GlycoPOST: GPST000201
Proteomics raw data	This paper	MassIVE: MSV000087640
Experimental models: cell lines		
Male T84 UAP1(F383G) cells	(Yu et al., 2012)	N/A
Recombinant DNA		
MGAT1-PGEn2	(Moremen et al., 2018) dnasu.org	DNASU: HsCD00413193
MGAT2-PGEn2	(Moremen et al., 2018) dnasu.org	DNASU: HsCD00413203
MGAT3-PGEn2	(Moremen et al., 2018) dnasu.org	DNASU: HsCD00413114
MGAT4A-PGEn2	(Moremen et al., 2018) dnasu.org	DNASU: HsCD00413158
MGAT5-PGEn2	(Moremen et al., 2018) dnasu.org	DNASU: HsCD00522293
POMGNT1-PGEn2	(Moremen et al., 2018) dnasu.org	DNASU: HsCD00413191
POMGNT2-PGEn2	(Moremen et al., 2018) dnasu.org	DNASU: HsCD00413126
B3GNT2-PGEn2	(Moremen et al., 2018) dnasu.org	DNASU: HsCD00413085
GCNT2B-PGEn2	(Moremen et al., 2018) dnasu.org	DNASU: HsCD00413202

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Software and algorithms		
MATLAB R2019a	MathWorks	https://www.mathworks.com/products/matlab.html
GraphPad Prism	GraphPad Software	https://www.graphpad.com/scientific-software/prism/
Byonic™ 2.3	Protein Metrics	https://proteinmetrics.com/byos/
SpectrsPicker	Dr. Rene Ranzinger	https://github.com/ReneRanzinger/SpectraFiltering/
Proteome Discover v2.2	Thermo Fisher Scientific	https://www.thermofisher.com/us/en/home/industrial/mass-spectrometry/liquid-chromatography-mass-spectrometry-lc-ms/lc-ms-software/multi-omics-data-analysis/proteome-discoverer-software.html

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Jennifer J. Kohler (jennifer.kohler@utsouthwestern.edu).

Materials availability

This study did not generate new unique reagents. Cell lines and limited amounts of Ac_3GlcNDAz-1P$(AcSATE)_2$ and UDP-GlcNDAz are available through the lead contact. The pSin4 UAP1(F383G) plasmid is available from Addgene using ID 169891. GlcNAc-transferase expression constructs are available from dnasu.org using the indicated clone IDs.

Data and code availability

Raw data of proteomics analysis were deposited to MassIVE with the identifier MSV000087640. Crosslinked bands from the first biological replicate from high, medial, and low molecular weight have the sample names f.MSV000087640/raw/LUM1_661503.raw, f.MSV000087640/raw/LUM1_661504.raw and f.MSV000087640/raw/LUM1_661505.raw, respectively. Control bands from the first biological replicate from high, medial, and low molecular weight have the sample names f.MSV000087640/raw/LUM1_661500.raw, f.MSV000087640/raw/LUM1_661501.raw, and f.MSV000087640/raw/LUM1_661502.raw, respectively. Crosslinked bands from the second biological replicate from high, medial, and low molecular weight have the sample names f.MSV000087640/raw/LUM1_671792.raw, f.MSV000087640/raw/LUM1_671790.raw and f.MSV000087640/raw/LUM1_671788.raw, respectively. Control bands from the second biological replicate from high, medial, and low molecular weight have the sample names f.MSV000087640/raw/LUM1_671793.raw, f.MSV000087640/raw/LUM1_671791.raw and f.MSV000087640/raw/LUM1_671789.raw, respectively.

The glycoproteomics data have been deposited at GlycoPost (Watanabe et al., 2021) with the dataset identifier GPST000201.

This paper does not report original code.

Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request

EXPERIMENTAL MODEL AND SUBJECT DETAILS

Male T84 cells stably overexpressing UAP1(F383G) were generated by infection with pSIN4 UAP1(F383G) lentivirus. (Rodriguez et al., 2015; Yu et al., 2012) Male T84 UAP1(F383G) cells (Yu et al., 2012) were cultured in DMEM/F12 medium (Thermo Fisher Scientific, Cat. No. 11330032) containing 5 % fetal bovine serum (Thermo Fisher Scientific, Cat. No. 16000044), 100 μg/mL streptomycin and 100 units/mL penicillin (Thermo Fisher Scientific, Cat. No. 15140122). Cells were free of mycoplasma contamination as confirmed by MycoAlert Detection Kit (Lonza, Basel, Switzerland; Cat# LT07-118). Cell line has not been authenticated.

METHOD DETAILS

Chemicals, antibodies, lectins, and other protein reagents

Dimethyl sulfoxide (DMSO) was purchased from Fischer Scientific (Cat. No. BP231-100). Ac_3GlcNDAz-1P$(AcSATE)_2$ and UDP-GlcNDAz disodium salt were synthesized by Wuxi AppTec (Tianjin). Stock concentration of Ac_3GlcNDAz-1P$(AcSATE)_2$ is 50 mM in DMSO. Lactose (Cat. No. 61339-25g), cellobiose (Cat. No. 22150-10G), D-galactose (Cat. No. G0750-100G), N-acetylglucosamine

(Cat. No. A3286-25G), D-glucose (Cat. No. G7021-100G), L-galactose (Cat. No. G7134-100MG), D-fucose (Cat. No. F8150-1G) and L-fucose (Cat. No. F2252-5G) were purchased from Sigma. Maltose (Cat. No. 190237) was purchased from MP Biomedicals. Sucrose was purchased from Fisher Scientific (Cat. No. BP220-1). Kifunensine (Cat. No. K1140-1MG) was purchased from Sigma and stock concentration is 1 mg/mL in Dulbecco's phosphate-buffered saline (DPBS). 2F-peracetyl-fucose (Cat. No. 344827-10MG) was purchased from Millipore and the stock concentration is 200 mM in DMSO. All stocks are stored at -20 ºC unless otherwise specified.

Antibodies were from the following sources: galectin-1 (clone D608T, #12936) (1:1000 for WB), LAMP1 (clone D2D11, #9091) (1:1000 for WB, 1:100 for IP), CDCP1 (clone D1W9N, #13794) (1:1000 for WB, 1:200 for IP), CDCP1(#4115) (1:1000 for WB) were purchased from Cell Signaling Technologies; cholera toxin B subunit (ab34992) (1:5000 for WB) and LY75 (ab124897) (1:2000 for WB, 1:50 for IP) were purchased from Abcam; α-tubulin (T6199) (1:2000 – 1:5000 for WB) was purchased from Sigma; CEA (clone 1106, MA5-14675) (1:2000 for WB, 1:100 for IP) was purchased from Thermo Fisher Scientific; galectin-1 (AF1152) (0.1 mg/mL stock, 1:500 dilution for WB used in comparison of crosslinking in intact cells and in lysates) was purchased from R&D Systems. IRDye® 680 RD conjugated goat anti mouse IgG (H+L) (926-68070), IRDye® 800CW conjugated goat anti rabbit IgG (H+L) (926-32211), IRDye® 800CW conjugated goat anti mouse IgG (H+L) (926-32210) and IRDye® 800CW conjugated donkey anti goat IgG (H+L) (925-32214) were purchased from LI-COR and used as 1:10000 dilution for WB.

Biotin-labeled lectins were purchased from Vector Laboratories: GSL II (B-1215), succinylated WGA (B-1025S), DSL (B-1185), LEL (B-1175), ECA (B-1145), SNA (B-1305), MAL II (B-1265), PHA-L (B-1115), ConA (B-1005). Unlabeled AAL (L-1390) was purchased from Vector Laboratories.

Recombinant galectin-1 and biotin-labeled galectin-1 were generated by Dr. Connie Arthur and Dr. Sean Stowell at Emory University.[ref] StcE was a gift from Drs. Stacy Malaker and Carolyn Bertozzi at Stanford University. (Malaker et al., 2019) Cholera toxin B subunit (C9903) was purchased from Sigma. Biotin-conjugated cholera toxin B subunit (C-34779) was purchased from Thermo Fisher Scientific. IRDye® 800CW conjugated streptavidin (926-32230) was purchased from LI-COR. Normal rabbit IgG (NI01-100UG) and normal mouse IgG (NI03-100UG) were purchased from Millipore Sigma.

Glycopeptide analysis of GlcNDAz incorporation into N-glycans

Cell culture. For analysis of GlcNDAz incorporation into N-glycans by mass spectrometry, 2.25 x 10^6 of T84 UAP1(F383G) cells were plated with 20 mL of complete medium in nine 15-cm tissue culture plates. After 24 h and 48 h, 40 μL of $Ac_3GlcNDAz\text{-}1P(AcSATE)_2$ stock or DMSO was added to each plate. About 72 hr after cells were plated, cells were lifted from the plates by incubation with 1 mM of ethylenediaminetetraacetic acid (EDTA) in DPBS at 37 ºC and quenched with cell culture medium. Cells were collected by centrifugation and washed with DPBS. Cell pellets were frozen in liquid nitrogen and stored at −80°C.

Enrichment of membrane proteins from the cells and protease digestion. For the membrane proteins enrichment, the T84 cells (30 X 10^6) were pelleted (W, whole cell lysate was prepared from an aliquot) at 1000 x g for 5 min and homogenized in high salt buffer (1.0 mL; 2.0 M NaCl, 5.0 mM EDTA, pH 7.4) by passing through 26-gauge syringe for 10 times. The homogenate was then probe sonicated for 5 min using 10 s intervals pulsed program and ultracentrifuged at 90,000 X g for 15 min at 4°C. The supernatant (S1) was separated and the pellet (P1) was re-suspended in 1.0 mL sodium carbonate buffer (0.1 M Na_2CO_3, 1.0 mM EDTA, pH 11.3) and kept for 30 min on ice. The suspension was subsequently ultracentrifuged at 90,000 x g for 90 min at 4°C. The supernatant (S2) and the pellet (P2) was separated and the pellet containing membrane fraction was dissolved in 1.0 mL urea lysis buffer (8.0 M urea, 1.0 M NaCl, 4% CHAPS, 100 mM dithiothreitol (DTT), 200 mM Tris-HCl, pH 8.0) and kept at 50°C for 45 min for denaturation. Subsequently, 55 mg of iodoacetamide was added and incubated at room temperature in dark for 45 min. The proteins were precipitated by sequential addition of chloroform (1.5 mL), methanol (4.5 mL) and water (5.0 mL) and the precipitated proteins were separated by centrifugation at 3000 X g for 5 min. The proteins were washed once with 500 μL of methanol, centrifuged, and resuspended in 50 mM ammonium bicarbonate buffer (pH 8.0).The proteins were digested by adding a cocktail of 25 μg each of trypsin-LysC protease and incubated at 37°C for 24 hr. MS compatible protease inhibitor cocktail was added to all buffers prior to addition.

Enrichment of glycopeptides from the protease digest. The glycopeptides were enriched by ZIC-HILIC method using ProteoExtract Glycopeptide Enrichment Kit (Millipore, 72,103-3) with some modification from manufacturer's protocol, as reported previously (Shajahan et al., 2020). About 100 μg of peptides (estimated by nanodrop spectrophotometer) in 30 μL were mixed with 150 μL of binding buffer. 150 μL of ZIC-HILIC resin was centrifuged to remove the solvents, the samples were added to it and vortexed for 20 min at room temperature. The samples with resin were centrifuged, supernatants were discarded, the resin was resuspended in 450 μL washing buffer and incubated at room temperature for 10 min. The washing procedure was repeated two more times. The enriched glycopeptides were eluted by adding 225 μL of elution buffer and subsequently by 225 μL of 0.1% formic acid, incubating for 5 min at room temperature in each case. The eluted fractions were combined, evaporated to dryness and the enriched glycopeptides were dissolved in 25 μL of 0.1% formic acid for the subsequent LC-MS/MS analysis.

Data acquisition of protein digest samples using nano-LC-MS/MS. The glycopeptides were analyzed on an Orbitrap Fusion Tribrid mass spectrometer coupled with a Dionex Ultimate 3000 LC-MS/MS system. Acclaim PepMap nano-LC columns (Thermo Scientific; Cat No. 164568) of 150 mm length with 75 μm internal diameter (id), filled with 3 μm, 100 ˚A C18 material (reverse phase) were used for chromatographic separation. Solvent A consisted of 0.1% formic acid (Fluka) in LC-MS grade water (Sigma Aldrich), and solvent B consisted of 0.1% formic acid, 80% acetonitrile (Sigma Aldrich) and 20% LC-MS grade water. Samples were run for 180 min at 300 nL/min with gradients as follows, 180 min LC method: 5-30% solvent B within 90 min; 30-60% solvent B within 60 min; 60-99% solvent B within 15 min; 99-80% solvent B within 5 min; 80-10% solvent B within 5 min and 10-5% solvent B within 5 min. All instrument methods for the mass spectrometer were set up in data dependent acquisition mode with an automatic gain

control (AGC) target value of 5 X 10^5 for precursor scans. After the precursor ion scan at 120,000 resolution in Orbitrap analyzer, intense precursors were selected for subsequent fragmentation using HCD and CID (product-triggered based on the presence of glycan oxonium ions in the HCD) within 3 s at a normalized collision energy of 28 and 35, respectively. For internal mass calibration 445.120025 ion was used as lock mass with a target lock mass abundance of 0%. Charge state screening and dynamic exclusion was enabled (exclusion size list 100, exclusion duration 30 s), and precursors with unknown charge state or a charge state of +1 were excluded. The fragment ions were analyzed in the Orbitrap analyzer at 15,000 resolution.

Identification of GlcNDAz modified glycopeptides. The identification of the LC-MS/MS glycopeptide spectra file was performed using the commercial software Byonic 2.3 (https://www.proteinmetrics.com/products/byonic/). In order to improve the detection of GlcNDAz modified N-glycopeptides, the LC-MS/MS data files were processed as described previously (Shajahan et al., 2020). Briefly, an in-house software SpectraPicker (freely available at our source code repository https://github.com/ReneRanzinger/SpectraFiltering/) that allows extracting MS scans based on user defined key ions in the MS^2 data is used to extract spectra with oxonium ions of GlcNDAz-N_2 (M + H - 244.12, +/− 5ppm accuracy). The spectra with an intensity value of more than 10% of the highest MS^2 intensity was saved into a separate filtered data file in mzXML format. Further, Byonic software-based search was conducted on the filtered spectra file for the annotation of the GlcNDAz bearing glycoproteins using a custom created glycan database with GlcNDAz modified N-glycans. The UniProtKB human reviewed *N*-glycoproteome proteome dataset (download date 14.10.2018) was used for the search with oxidation of methionine and carbamidomethylation of cysteine as variable modifications. A precursor ion tolerance of 5 ppm and fragment ion tolerance of 15 ppm was set for the search with up to two missed cleavage for the target enzyme trypsin/Lys-C.

In vitro GlcNAc transferase activity assay

Protein expression constructs encoding the catalytic domains of MGAT1, MGAT2, MGAT3, MGAT4A, MGAT5, POMGNT1, POMGNT2, B3GNT2, GCNT2B were generated by PCR from a Mammalian Gene Collection clone followed by Gateway recombination into the pDONR221 vector (Moremen et al., 2018). The PCR amplification extended the truncated protein coding regions by inclusion of flanking Gateway att1 recombination sites as well as an extension of the NH_2- terminus of the coding region with a TEV protease recognition site as previously described (Moremen et al., 2018). The fusion protein constructs encode an NH_2-terminal signal sequence, 8xHis tag, AviTag, ''superfolder'' GFP, the TEV protease recognition site, and the corresponding truncated protein coding region behind a CMV promoter. Recombinant enzyme expression was accomplished by transient transfection of HEK293 cells (FreeStyle 293-F cells, Life Technologies, Grand Island, NY) as previously described (Moremen et al., 2018). Briefly, HEK293 cells were maintained in serum free Freestyle 293 expression medium (Life Technologies). Transfections were initiated at cell densities of 2.5 × 10^6 cells/ml by addition of 4 μg/mL of the respective expression plasmid and 9 μg/mL polyethylenimine (linear 25 kDa PEI, Polysciences, Inc., Warrington, PA) to the suspension culture. The cultures were diluted 1:1 with culture medium containing 4.4 mM valproic acid (2.2 mM final) 24 hr after transfection, and protein production was continued for a further 5 days at 37°C.

Culture medium was harvested on day 6 and clarified by sequential centrifugation at 1,500 rpm for 10 min, 2,500 rpm for 15 min, 4,000 rpm for 20 min and then passed through a 0.8 μM filter (Pall Corporation). The conditioned culture medium was loaded on a Ni^{2+}-NTA Superflow (Qiagen) column equilibrated with 20 mM HEPES, 300 mM NaCl, 20 mM imidazole, pH 7.4, washed with column buffer, and eluted successively with column buffers containing stepwise increasing imidazole concentrations (40-300 mM). The protein preparations were further purified on a Superdex 75 column (GE Healthcare) preconditioned with a buffer containing 20 mM HEPES, 300 mM NaCl and the final purified protein was concentrated by ultrafiltration to 1 mg/mL.

Enzyme activity was determined using the UDP-Glo Glycosyltransferase Assay (Promega) that determined UDP concentration formed as a by-product of the GlcNAc transferase reaction. Assays were performed according to the manufacturer's instructions using reactions (5 μL) that consisted of a universal buffer containing 100 mM each of MES, MOPS, TRIS, pH 7.0, 0.5 mM donor (UDP-GlcNDAz and UDP-GlcNAc), 0.5 mM acceptor, 5 mg/mL BSA, 5 mM $MnCl_2$ (for MGAT1, MGAT2, MAGT3, MGAT4A, POMGNT1, B3GNT2), and purified enzymes (1-500 ng). Acceptor structures for the respective enzymes were: $Man_3GlcNAc_2$-Asn (MGAT1), $GlcNAcMan_3GlcNAc_2$-Asn (MGAT2), $GlcNAc_2Man_3GlcNAc_2$-Asn (MGAT3, MGAT4A, MGAT5) (all kind gifts of Geert-Jan Boons, University of Georgia), O-Man glycopeptide 317 (POMGNT1 and POMGNT2) (kind gift of Lance Wells, University of Georgia), lacto-N-neotetraose (B3GNT2 and GCNT2B) (Carbosynth). Assays were initiated with the addition of enzyme and carried out for 1 hr at 37°C. Reactions were stopped by mixing with an equal volume of UDP Detection Reagent (5 μL) in white polystyrene, low-volume, 384-well assay plates (Corning) and incubated for 60 min at room temperature. After incubation, luminescence measurements were performed using a GloMax Multi Detection System plate reader (Promega). Luminescence values were compared to a standard curve for quantification of UDP produced.

Lectin binding

Cell culture. To evaluate the global cellular glycome changes resulting from GlcNDAz incorporation, 500,000 T84 UAP1(F383G) cells were plated with 5 mL of complete medium in a 6-cm tissue culture plate. At 24 hr and 48 hr after cells were plated, 10 μL of $Ac_3GlcNDAz$-1P$(AcSATE)_2$ stock or DMSO was added to each plate. About 72 hr after cells were plated, cells were rinsed with DPBS and scraped into 500 μL of RIPA buffer (50 mM Tris-HCl, pH = 8.0, 150 mM NaCl, 0.1% (w/v) SDS, 0.5% (w/v) sodium deoxycholate and 1% (w/v) IGEPAL CA-630) with protease inhibitor cocktail (Sigma, Cat. No. 11836170001). Cells were lysed by

vortexing followed by incubation on ice for 30 min. Lysates were centrifuged at 20,817 *g* at 4°C for 10 min. Concentrations of proteins were determined using a Pierce BCA Protein Assay Kit (Thermo Fisher Scientific, Cat. No. 23225) with BSA as standard. Supernatant was frozen in liquid nitrogen and stored at −80°C.

Lectin binding. For lectin blots, samples were prepared by dilution of lysates with water and addition of 4 x SDS loading dye (200 mM Tris-HCl pH 6.8, 8% (w/v) SDS, 0.08% (w/v) bromophenol blue, 40% (v/v) glycerol, and 40 mM DTT). Samples were denatured at 90°C for 5 min and 10 μg of lysates were loaded on each lane and separated by SDS-PAGE using a 12% TGX FastCast gel (Bio-Rad, Cat. No. 1610175). Proteins were transferred to a Immobilon-FL PVDF membrane (Millipore, Cat. No. IPFL00010 or IPFL00005) by wet transfer with 87 mA of current at 4°C overnight. After the transfer, the membrane was rinsed with ultrapure water and stained with a Revert 700 Total Protein Stain Kits (LI-COR, Cat. No. 926-11016 or 926-11,010). Total protein stain was imaged with a ChemiDoc MP Imaging system (Bio-Rad, Hercules, CA). Membrane was destained with the Revert 700 Total Protein Stain Kits, rinsed with ultrapure water, and equilibrated with TBS (10 mM Tris, pH 8.0, 150 mM NaCl) for about 2 min at room temperature. Blots were blocked with Carbo-free Blocking Solution (Vector Laboratories, SP-5040) diluted in TBS and then probed with 10 μg/mL of biotin-labeled lectins or galectin-1, or 1 μg/mL of biotin-labeled cholera toxin B subunit diluted in Carbo-free Blocking Solution in TBST (10 mM Tris, pH 8.0, 150 mM NaCl, 0.1% Tween 20) at 4°C overnight. Blots were washed with TBST and probed with IRDye 800CW conjugated streptavidin at 1:4000 dilution in Carbo-free Blocking Solution in TBST with 0.02% SDS for 1 hr at room temperature. Blots were washed with TBST and imaged on a ChemiDoc MP Imaging system.

On-Cell ELLA. To each well in a 96-well stripwell plate (Costar, 9102), 25,000 T84 UAP1(F383G) cells were added in 200 μL of complete medium. At 24 hr and 48 hr after cells were plated, medium was replaced with complete medium containing 100 μM of $Ac_3GlcNDAz\text{-}1P(AcSATE)_2$ or equal volume of DMSO. About 72 hr after cells were plated, cells were rinsed three times with DPBS and fixed with 4% paraformaldehyde (PFA, Electron Microscopy Sciences, Cat. No. 15713) for 30 min at room temperature. Cells were then washed with DPBS. DPBS supplemented with 2% (w/v) BSA and 0.1% (w/v) lysine were added to cells and incubated for 1 hr at room temperature to quench unreacted PFA and block for nonspecific binding. Cells were then incubated with biotin-labeled lectins diluted in 2% BSA in DPBS overnight at 4°C and washed three times with DPBS. Cells were then incubated with 1:4000 dilution of streptavidin-HRP (500 U/ml, Roche, Cat. No. 11089153001) in 2% BSA in DPBS for 1 hr at room temperature and washed three times with DPBS. To evaluate the activity of HRP, HRP solution was prepared by dissolving o-phenylenediamine dihydrate (OPD) (Sigma, Cat. No. P1526-10G) in HRP buffer (2.56 g citric acid and 3.65 g anhydrous dibasic sodium phosphate in 500 mL of water) to a final concentration of (w/v) 0.1% and 30% (w/w) H_2O_2 was added at a 1:1000 dilution. Reaction was started by addition of 100 μL of HRP solution to each well and quenched with 100 μL of 1 M H_2SO_4. Absorbance at 490 nm was determined with a Synergy 2 Multi-Mode Microplate Reader (BioTek, Winooski, VT) and corrected with absorbance at 650 nm. Cells were then washed and incubated with BCA reagents for 2 hr at 37°C. Absorbance at 562 nm was determined, corrected with the average absorbance of blank 96-well plate and used to normalize the readout of HRP activity to total protein concentration.

For each biological replicate, three technical replicates were included. The average of lectin binding from three technical replicates was normalized to the highest value in DMSO-treated cells at the following lectin concentrations: 30 μg/mL of lectins for GSL II, SNA, MAL II and CTB; 1 μg/mL of lectins for succinylated WGA and ECA; 3 μg/mL of lectins for DSL and LEL; 10 μg/mL of lectins for PHA-L and galectin-1. The normalized average lectin binding from four biological replicates were used for statistical analysis using Student's t-test and plotting.

Crosslinking of galectin-1 and CTB to glycoproteins

To crosslink galectin-1 and CTB to glycoproteins, 150,000 T84 UAP1(F383G) cells were plated in 1 mL of complete media in each well in 12-well plates. To each well, 2 μL of $Ac_3GlcNDAz\text{-}1P(AcSATE)_2$ stock or DMSO was added at 24 hr and 48 hr after cells were plated. At about 72 hr after cells were plated, cells were rinsed with DPBS and changed to 500 μL of serum-free media. Galectin-1 or CTB was added to cells at a final concentration of 10 μg/mL, or 4 μg/mL, respectively. Cells were incubated with galectin-1 or CTB for 45 min at 4°C in dark and irradiated with a UVP Blak-Ray Lamp (Model XX-20BLB, VWR, Cat. No. 21474-676) for 45 min on an ice-water bath. Control cells were incubated with galectin-1 or CTB for 90 min at 4°C in dark. After UV irradiation, cells were scraped into 80 μL of RIPA buffer with protease inhibitor cocktail. Cells were lysed and protein concentrations were determined as described for the lectin blots.

For immunoblots, 30 μg of lysates were loaded on each lane and separated by SDS-PAGE using a 7.5% TGX FastCast Gel (BioRad, 1610171). Proteins were transferred to Immobilon-FL PVDF membrane with 87 mA of current at 4°C overnight and stained for total protein as described for lectin blots. After destaining, blots were rinsed with ultrapure water, equilibrated with TBS and blocked with a Intercept (TBS) blocking buffer (LI-COR, 927-66003) for 1 hr at room temperature. Blots were then incubated with anti-galectin-1 or anti-CTB and anti-α-tubulin antibodies diluted in a Intercept T20 (TBS) antibody diluent (LI-COR, 927-66003) at 4°C overnight. Blots were washed with TBST and incubated with IRDye 800CW conjugated goat anti rabbit IgG (H + L) and IRDye 680 RD conjugated goat anti mouse IgG (H + L) diluted in Intercept T20 (TBS) antibodies diluent containing 0.01% SDS for 1 hr at room temperature. Blots were then washed with TBST and imaged with a ChemiDoc MP Imaging system.

For experiments in which inhibitors of glycosylation were used, 200 μM of 2F-peracetyl-fucose or 1 μg/mL of kifunensine was included in the cell culture media when cells were plated. Cells were treated with $Ac_3GlcNDAz\text{-}1P(AcSATE)_2$ stock or DMSO, crosslinked to galectin-1 or CTB and analyzed by immunoblot as described above.

To evaluate inhibition of crosslinking by free sugars, lactose, maltose, sucrose, cellobiose, D-galactose, D-glucose, L-galactose, D-GlcNAc, D-fucose and L-fucose were dissolved in serum-free media to a final concentration of 100 mM and filtered through a

0.2 μm Nalgene syringe filter (Thermo Fisher Scientific, Cat. No. 720-1320). T84 UAP1(F383G) cells were cultured as described above. Serum-free media (500 μL) with free sugars was added to each well. Galectin-1 or CTB was then immediately added and UV irradiation was performed. Cells were lysed and analyzed by immunoblot as described above.

To evaluate inhibition of crosslinking by lectins, T84 UAP1(F383G) cells were cultured as described above and changed to 500 μL of serum-free media. Biotin-labeled lectins were added to cells to a final concentration of 40 μg/mL immediately before the addition of galectin-1 or CTB. UV irradiation was performed and lysates were analyzed by immunoblot as described above.

To remove O-glycans on mucin or mucin-like proteins with StcE, cells were washed with DPBS and incubated with 5 μg/mL of StcE in serum-free media at 37°C for 2 hr before adding galectin-1 and performing the crosslinking experiment.

Proteomic analysis of galectin-1 crosslinked complexes

Cell culture and crosslinking. To prepare crosslinked samples for proteomic analysis, 2 x 10^6 T84 UAP1(F383G) cells were plated in 10 mL of media in a 10 cm tissue culture plate. At 24 hr and 48 hr after cells were plated, 20 μL of $Ac_3GlcNDAz\text{-}1P(AcSATE)_2$ stock or DMSO was added to each plate. About 72 hr after cells were plated, cells were rinsed with DPBS and 3 mL of serum-free media was added to each plate. Biotin-labeled galectin-1 was added to serum-free media to a final concentration of 100 μg/mL. Cells treated with $Ac_3GlcNDAz\text{-}1P(AcSATE)_2$ were incubated with galectin-1 for 45 min at 4°C and UV irradiated for 45 min on ice. Cells treated with DMSO were kept at 4°C in dark for 90 min without galectin-1. Cells were then washed twice with 5 mL of 100 mM lactose in cold DPBS followed by 10 mL of DPBS to remove uncrosslinked galectin-1. After washing, cells were scraped into 1 mL of RIPA buffer with protease inhibitor cocktail, vortexed, and incubated on ice for 30 min. Lysates were centrifuged at 20,817 *g* for 10 min at 4°C. Concentration of supernatant was determined with a Pierce BCA Protein Assay Kit. Lysates were frozen in liquid nitrogen and stored at −80°C.

Immunoprecipitation and silver stain. To purify the crosslinked complexes, 1043 μg of lysates were diluted with RIPA buffer containing protease inhibitor cocktail to 1 mL and added to 10 mg of Dynabeads M-280 streptavidin (Thermo Fisher Scientific, Cat. No. 11206D), which had been pre-equilibrated with RIPA buffer. Lysates were incubated with beads at 4°C overnight with end-over-end rotation. Beads were washed 5 times with 1 mL of RIPA buffer and 5 times with 1 mL of high salt RIPA buffer (50 mM Tris-HCl, pH 8.0, 500 mM NaCl, 0.1% (w/v) SDS, 0.5% (w/v) sodium deoxycholate and 1% (w/v) IGEPAL CA-630). After washing, beads were transferred to a fresh tube and eluted by incubating beads with 100 μL of 1 x SDS dye with biotin (50 mM Tris-HCl, pH 6.8, 2% (w/v) SDS, 0.02% (w/v) bromophenol blue, 10% (v/v) glycerol, 10 mM DTT and 20 mM biotin) at 90°C for 5 min. Eluent from each sample was loaded in 2 lanes of a 7.5% precast gel (Bio-Rad, Cat. No. 4561024) for separation by SDS-PAGE. After separation, gels were stained with a mass compatible silver staining kit (Thermo Fisher Scientific, Cat. No. 24600).

In-gel digestion and mass spectrometry. Gel bands were digested overnight with trypsin (Pierce) following destaining, reduction with DTT and alkylation with iodoacetamide (Sigma–Aldrich). The samples then underwent solid-phase extraction cleanup with an Oasis HLB plate (Waters) and the resulting samples were injected onto an Orbitrap Fusion Lumos mass spectrometer coupled to an Ultimate 3000 RSLC-Nano liquid chromatography system. Samples were injected onto a 75 um i.d., 75-cm long EasySpray column (Thermo) and eluted with a gradient from 0 to 28% buffer B over 90 min. Buffer A contained 2% (v/v) ACN and 0.1% formic acid in water, and buffer B contained 80% (v/v) ACN, 10% (v/v) trifluoroethanol, and 0.1% formic acid in water. The mass spectrometer operated in positive ion mode with a source voltage of 1.5–2.2 kV and an ion transfer tube temperature of 275°C. MS scans were acquired at 120,000 resolution in the Orbitrap and up to 10 MS/MS spectra were obtained in the ion trap for each full spectrum acquired using higher-energy collisional dissociation (HCD) for ions with charges 2-7. Dynamic exclusion was set for 20 s after an ion was selected for fragmentation.

Raw MS data files were analyzed using Proteome Discoverer v2.2 (Thermo), with peptide identification performed using Sequest HT searching against the human protein database from UniProt. Fragment and precursor tolerances of 10 ppm and 0.6 Da were specified, and three missed cleavages were allowed. Carbamidomethylation of Cys was set as a fixed modification, with oxidation of Met set as a variable modification. The false-discovery rate (FDR) cutoff was 1% for all peptides.

Immunoprecipitation

Cell culture and crosslinking. To prepare crosslinked samples for immunoprecipitation, T84 UAP1(F383G) cells were treated, crosslinked and harvested as sample preparation for proteomics, except that 30 or 100 μg/mL of unlabeled galectin-1 was used during crosslinking, and that 100 mM lactose was added to one sample right before the addition of galectin-1 as a negative control.

In-lysate crosslinking. To compare the crosslinking of LAMP1 and CEA to galectin-1 in intact cells and in lysates, T84 UAP1(F383G) cells were plated and treated as for proteomics. For crosslinking in lysates without lactose, cells were scraped into 1 mL of RIPA buffer with a protease inhibitor cocktail, vortexed and incubated on ice for 30 min. Lysates were then centrifuged at 20,817 *g* for 10 min at 4°C and 900 μL of the supernatant from each sample was transferred to one well in a 6-well plate. For crosslinking in lysates with the presence of lactose, cells were lysed as before, except that RIPA buffer with 100 mM of lactose and protease inhibitor cocktail was used to lyse the cells. Unlabeled galectin-1 was added to intact T84 cells in serum-free media or to cell lysates to a final concentration of 30 μg/mL, incubated for 45 min at 4°C in dark and UV-irradiated for 45 min on an ice-water bath. Intact T84 cells were then lysed as before. Crosslinked lysates were directly transferred to a fresh 1.5 mL tube, incubated on ice for 30 min and centrifuged at 20,817 *g* for 10 min at 4°C. Protein concentration was determined with the BCA kit using RIPA buffer or RIPA buffer with 100 mM lactose as blank.

Immunoprecipitation and Western blot. To enrich for selected proteins, 140 μg of lysates were diluted to 1 mg/mL in RIPA buffer with a protease inhibitor cocktail and incubated with LAMP1, CEA, CDCP1 and LY75 antibody or the same amount of mouse or rabbit IgG control overnight in dark at 4°C with end-over-end rotation. For 140 μg of lysates, 28 μL of Dynabeads Protein G (Thermo Fisher Scientific, Cat. No. 10004D) pre-equilibrated with RIPA buffer were added and incubated with lysates at room temperature with end-over-end rotation for 1 hr. Beads were washed 4 times with 200 μL of RIPA buffer and eluted with 28 μL of 1 x SDS loading dye (50 mM Tris-HCl, pH 6.8, 2% (w/v) SDS, 0.02% (w/v) bromophenol blue, 10% (v/v) glycerol and 10 mM DTT) at 90°C for 5 min. For IP control (IP and probe for the same protein), 4 μL of the eluent was loaded and separated on a 7.5% SDS PAGE. For IP experiment (IP for potential galectin-1 binding partner and probe for galectin-1), 20 μL of the eluent was loaded and separated on a 7.5% SDS PAGE. Proteins were then transferred to PVDF-FL membrane and membrane was blocked same as for detection of galectin-1 cross-linking. After blocking, membrane was probed with primary antibody overnight at 4°C, washed with TBST and incubated with IRDye conjugated secondary antibody for 1 hr at room temperature. Blots were imaged with a ChemiDoc MP Imaging system, after which the region below around 50 kDa was removed. To enhance the signal to noise ratio, blots were then probed again with primary antibody overnight at 4°C, washed with TBST and incubated with IRDye conjugated secondary antibody for 1 hr at room temperature, and imaged again with ChemiDoc MP Imaging system.

High performance anion exchange chromatography (HPAEC) analysis of UDP-GlcNDAz

To assess production of UDP-GlcNDAz, 2 x 10^6 T84 UAP1(F383G) cells were plated in a 6-cm tissue culture plate and allowed to adhere to plates overnight. Cells were treated with DMSO or Ac_3GlcNDAz-1P$(AcSATE)_2$ at a final concentration of 100 μM for 8 hr. Cells were lifted from plates by TrypLE (Thermo Fisher Scientific, Cat. No. 12605-010), quenched with complete medium and washed with DPBS. Cells were frozen in liquid nitrogen and stored at −80°C.

To extract the nucleotide sugars, cells were resuspended in 1 mL of ice-cold 50% acetonitrile, vortexed, and incubated on ice for 10 min. Debris was removed by centrifugation at 20,000 *g* at 4°C for 10 min. Supernatant was then frozen in liquid nitrogen and dried with a SAVANT SC210A SpeedVac Concentrator (Thermo Fisher Scientific).

Before analysis by HPAEC, extract was resuspended in 100 μL of 40 mM sodium phosphate buffer (pH 7.4) and centrifuged at 20,817 *g* at 4°C for 1 min. The supernatant was filtered through a 3K Amicon Ultra centrifugal filter unit at 4°C (Millipore, Cat. No. UFC5003BK), and 20 μL of each sample was injected for analysis by HPAEC using a CarboPac PA1 column (Thermo Fisher Scientific, Cat. No. 035391) with a guard column (Thermo Fisher Scientific, Cat. No. 0343096). Samples were eluted with buffer A (1 mM NaOH) and buffer B (1 M NaOAc, 1 mM NaOH) at a flow rate of 1 mL/min using either of the following gradient of buffer B in buffer A: T 0 min = 5%, T 40 min = 30%, T 45 min = 50%, T 60 min = 55%, T 65 min = 100%, T 75 min = 100%, T 80 min = 5%, T 90 min = 5%; or T 0 min = 20%, T 10 min = 45%, T 25 min = 45%, T 35 min = 55%, T 40 min = 100%, T 50 min = 100%, T 55 min = 20%, T 65 min = 20%. Elution of nucleotide sugars were monitored by UV absorption at the wavelength of 260 nm. UDP-GlcNDAz and UDP-GlcNAc (Promega, Cat. No. V7071) were used as standards.

QUANTIFICATION AND STATISTICAL ANALYSIS

Statistical analysis of lectin binding using ELLA was performed using two-tailed unpaired t test at concentrations indicated in the figure legend. A significant difference is defined when $p < 0.05$. ELLA data are from four biological replicates, each of which is averaged from three technical replicates, and error bars represent ±SD. Details of data acquisition can be found in the methods details. All Western blots and immunoprecipitation data are representative of three biological replicates, except for immunoprecipitation experiment to confirm proteomics hits, where data are representative of two biological replicates using 100 ug/mL galectin-1 and similar results have been obtained using 30 ug/mL galectin-1 from one trial. Proteomics results are averaged from two biological replicates. Glycoproteomics data are from single trial due to discontinuation of the ZIC-HILIC enrichment kit.